

Peer Review File

AI hybrid survival assessment for advanced heart failure
patients with renal dysfunction



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REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

The study by Zhang and colleagues aims to address the issue of limited value of existing algorithms in prognostication of patients with advanced heart failure (AHF) with renal dysfunction (RD). The authors applied AI algorithms to medical records from the Henan Province Clinical Research Center for Cardiovascular Diseases and validated on medical records available from the Beth Israel Deaconess Medical Center. Multiple AI/ML approaches were utilized, and the workflow integrated to establish a survival assessment algorithm that was tested in the validation cohort and compared to other predictive algorithms and found to be comparable or better. The model was also deployed on the web.

Could the authors clarify about the Henan database – are these all hospitalization records or both inpatient and outpatient records?

I have some concerns about the definition of AHF: inoperable severe congenital anomalies or severe valvular anomalies; isolated right heart insufficiency; severe LV structural abnormalities or diastolic dysfunction defined by HFpEF.

Can the authors clarify the thresholds for persistence of elevated BNP or NT-proBNP – I assume they were different for HFrEF and HFpEF.

The Henan cohort seems to be predominantly HFpEF with a surprisingly lower (<20%) HFrEF. Would it be better to focus on a specific type of heart failure since AHF may be different in terms of prognosis and treatment options in HFpEF and HFrEF?

Missing values – could the authors state how much data was missing?

Reviewer #2 (Remarks to the Author):

The submitted manuscript discusses an important issue: predicting advanced heart failure in patients with renal dysfunction. The authors have developed and implemented an AI-based tool that offers improved accuracy and accessibility compared to the available risk scores. However, there are some issues that should be clarified in order to fully appreciate the comprehensive methodology the authors have used:

- As the authors have developed a prediction system, it would be beneficial if they could provide the TRIPOD statement and checklist.
- When defining a study outcome, it would be helpful if the authors could provide more details. Why have they used all-cause mortality and not a specific cause, at least for a sensitivity analysis? What particular length of survival are the authors looking for? Later on, they report survival at 6/12/24/30 months, and presumably, that's what should be specified.
- It is not entirely clear which particular outcome has been used to implement a consensus assessment strategy. Was it all-cause mortality across the whole time span? Which was then assessed in various subgroups; is that correct?
- With regards to the data, it would be beneficial if the authors could provide more details on the imputation. Is the missing at random mechanism justified? How much data was missing, and for which parameters? How did the missingness vary across the training and testing sets? With the biomarkers, for example, frequently the data might be "missing" depending on the patient's state - those who are "healthier" would have fewer examinations. Please add a bit more detail to that.
- The average age in the external cohort is much higher than in the training cohort. Also, the all-cause mortality is much higher. Could the authors expand on that, please, and discuss why they believe this should not impact the generalisability of the results?

Addressing the issues above would allow for a full assessment of the work that the authors have performed."

These revisions improve the clarity and readability of the text.

Reviewer #3 (Remarks to the Author):

This work developed an effective survival prediction model by integrating diverse features from medical records as well as clinical markers using 12 machine learning methods for heart failure with renal dysfunction. They demonstrated the outperformance of the proposed model/system CPS in both internal and external clinical cohorts when compared with existing survival prediction tools. My comments as follows:

The key standing point of the study is there is a lack of composite approaches with machine learning for survival prediction in AHF patients with RD (line 83). However, there is little that has been discussed on why standalone studies are needed here, rather than using approaches for AHF or other similar diseases (there is a wide range of prediction models), despite general challenges for survival prediction in the clinic were mentioned (lines 85-92). The key references 8 and 9 are somewhat outdated as well. So there appears to be a gap in the rationale for the motivation behind the study, specifically regarding the clinical evidence motivating this study for AHF with RD.

The process of CPS system development (lines 157-203) isn't quite clear to me. A graphic illustration would be very helpful, e.g. how features are initially selected as well as for each ML model, how were they combined, which set of data was used at each training/validation step. Given this is one of the key contributions of the study, a more organized and clear presentation is essential in the Method.

When comparing the performance of CPS with other published scores (lines 311-320), it's not clear whether the published scores/models from the referenced study or scores/models re-trained on the training data of the current study using the approach from the referenced study were applied here. If it's the former, is there any concern of bias in the clinic/cohort

setting when applying these existing scores/models? Also, it's not clear to me whether the outperformance of CPS is because of the addition of the new input features (the 48 selected markers in CPS versus the input markers/features used in scores compared with; or because of the additional features from hospital records versus clinical markers only), or the modelling of the complex relationships between these features? Would be good to compare the markers/features input into each of these models, as well as the statistical/modelling approach used.

This site is not accessible <http://www.chictr.org>, so we won't be able to track important aspects of the study, e.g. ethnics.

Text sizes in some figures are too small (or low resolution) to read, e.g. Figure S1 and Figure S2. So I can't read these results and make judgements.

Legends of most figures only give a title without giving the details. Very difficult to track what the plots are exactly showing.

Point to Point Response

Manuscript Number: NCOMMS-23-44173A

Title: **Derivation and validation of an AI hybrid survival assessment system for advanced heart failure with renal dysfunction: a retrospective multicenter cohort study**

Reviewer's comment: Black

Authors' responses: Blue

Revised text in the re-submitted files: Red

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

The study by Zhang and colleagues aims to address the issue of limited value of existing algorithms in prognostication of patients with advanced heart failure (AHF) with renal dysfunction (RD). The authors applied AI algorithms to medical records from the Henan Province Clinical Research Center for Cardiovascular Diseases and validated on medical records available from the Beth Israel Deaconess Medical Center. Multiple AI/ML approaches were utilized, and the workflow integrated to establish a survival assessment algorithm that was tested in the validation cohort and compared to other predictive algorithms and found to be comparable or better. The model was also deployed on the web.

Response:

Thank you very much for your insightful recommendations. We are sure your advice will significantly enhance the rigor, reliability, and meaningfulness of our study. As you mentioned later, our manuscript has some pending issues, such as clarification of the database situation, details of the eligibility criteria for participants, discussion of the baseline characteristics of the population cohort, and the missing value situation.

To address this gap, we have conducted more detailed data analyses, and further elaborated and discussed our study process and results in more detail to refine the development and validation process of our modeling system. This has led to a more comprehensive understanding of the clinical implications of our findings. We have also embellished and restructured the manuscript to improve its readability and logical flow. Detailed information and raw data related to the analysis results have been added to enhance the reproducibility of the research data, and are now included in the **Revised Supplementary Tables 1-20**. Every single figure and Supplementary figure have been polished and restructured for clarity and readability in the **Revised Figures 1-7** and **Revised Supplementary Figures 1-10**. We sincerely hope that these revisions will meet your approval!

To begin with, we would like to provide a clear overview of the revised manuscript and research:

This study analyzed electronic health records of 1736 AHF&RD participants who met the strict inclusion and exclusion criteria, encompassing an in-house cohort (Henan Province Clinical Research Center for Cardiovascular Diseases encompassing 11 hospital subcenters) and an independent external cohort (Beth Israel Deaconess Medical Center).

(1). Population characteristics overview

In the result subsection 1, we detailed and discussed the demographic and clinical characteristics of both cohorts. Our trajectory analysis has revealed that the development and progression of AHF&RD does not follow a straightforward, linear path; rather, it is characterized by significant, nonlinear shifts that underscored the intricate and dynamic nature of disease evolution in advanced heart failure combined with renal dysfunction.

(2). Survival assessment system: AIHFLevel

In subsection 2, we leveraged our AI hybrid modeling framework, integrating the use of hybrid learners, single learner, and enhanced existing learners, which could minimize generalization error and improve the reliability of predictive outcomes. We selected the optimal modeling scheme from 132 candidates through a comprehensive evaluation process, resulting in establishing our survival assessment system **AIHFlevel (previously named CPS)**. This strategy mitigated the issue of overfitting as well as the impact of researcher bias to some extent.

(3). Uniformly prognostic implication and predictive performance of AIHFLevel

In subsection 3, we employed various methods, including restricted cubic spline analysis, univariate Cox regression analysis, Kaplan-Meier curves, ROC analysis, calibration analyses, and decision curve analysis, to demonstrate the consistent predictive accuracy and clinical utility of AIHFLevel across the discovery, replication, and meta cohorts.

(4). Robustness and superior performance of AIHFLevel

In subsection 4, we aimed to compare the performance of AIHFLevel with existing tools.

i. **Clinical traits:** In clinical practice, prognostication and stratification for management traditionally relied on a range of clinicopathological characteristics, such as serum cardiac troponin levels, natriuretic peptides, pharmacotherapy, renal function, age, and comorbidity status. AIHFLevel consistently outperformed 93 readily accessible clinical traits in predictive accuracy across the discovery, replication, and meta cohorts.

ii. **Composite prognostic models:**

(a) Risk scores were computed for each model based on their predefined features and coefficients as outlined in their original publications. AIHFLevel consistently exhibited superior accuracy compared to other models across cohorts.

(b) AIHFLevel and other models were retrained using only their respective predictors—without incorporating their original coefficients—through proportional hazards regression. AIHFLevel and AIHFLevel (re-fit) consistently exhibited superior accuracy compared to other models (re-fit) across cohorts.

(c) Other models were retrained using only their respective predictors—without incorporating their original coefficients—through Surv.Xgboost algorithm (a key component in AIHFLevel's development). Once again, AIHFLevel exhibited superior accuracy compared to other models (re-fit) across cohorts.

(5). Clinical interpretability underlying AIHFLevel

In subsection 5, we initially confirmed AIHFLevel's significant independent prognostic value through the multivariate Cox regression analysis and subgroup analysis.

We leveraged Shapley Additive exPlanation approach to endow AIHFLevel with both a global explanation that describes the overall functionality of a model and a local explanation that details how a certain prediction is made for an individual child by inputting the individualized data.

(6). Prognostic stratification underlying AIHFLevel

In subsection 6, we aimed to uncover the heterogeneity in trends for prognosis among AHF&RD patients. The conditional inference survival tree equipped the AIHFLevel system with a prognostic stratification function, categorizing individuals into three prognostic states (High-risk, Intermediate-risk, Low-risk) and identifying a probability threshold for stratification. The robustness of our stratification was confirmed through TrinROC analysis, Kaplan–Meier analysis, t-SNE dimensional reduction technique, and chi-square test.

We further validated the clinical relevance and the rationality of our prognostic stratification, suggesting its potential in guiding early preventive strategies through a tipping-point theory-based model, the enhanced dynamic network biomarker (DNB), and Euclidean distance-derived global variance analysis.

(7). Sensitivity analysis

AIHFLevel was initially trained with ACM as its primary endpoint. In subsection 7, when evaluated against MACE as an alternative endpoint, it still demonstrated reliable predictive performance. The ability to accurately assess outcomes across different clinical endpoints

indicated AIHFLevel's comprehensive predictive capabilities and robustness in clinical prognostication.

(8). Extrapolation of AIHFLevel to the heterogeneous populations

AIHFLevel has undergone rigorous evaluation within our designated testing datasets (Replication and Meta cohorts). However, external validation remains crucial. Models may erroneously correspond or be overfitted to peculiarities in training datasets, leading to a loss of generalizability to heterogeneous data unseen during the training process. The imperative for future AI systems lies in their broad applicability across diverse healthcare settings and geographic locations. In subsection 8, the significant population heterogeneity observed between CRCCD and BIDMC centers provided a robust test bed for assessing AIHFLevel's generalizability. The commendable performance of AIHFLevel in external validations highlighted its robust extrapolation and generalizability, indicating its potential as a clinically translatable tool.

(9). Convenient web application for clinical utility

In subsection 9, the AIHFLevel system has been deployed as a user-friendly application (<https://www.hf-ai-survival.com>) to enhance its utility in clinical scenarios. The system offers a comprehensive range of functionalities for individual patient analysis, including prognostic stratification, estimation of long-term survival probabilities, predictions of all-cause mortality at specific time points, local interpretation, and assessment of predictor contributions.

Overall, aligning values of risk predictors (as indicated by the AIHFLevel system) closer to their normal ranges through targeted interventions or management strategies could mitigate the risk of all-cause mortality (ACM) for this individual, even if the overall prediction may be pushing this case into the high-risk prognosis. This system could help physicians select the optimal treatments and offer personalized recommendations to improve outcomes based on the output risk stratification and interpretability of personal information.

Q1: Could the authors clarify about the Henan database – are these all hospitalization records or both inpatient and outpatient records?

Response:

We greatly appreciate your kind and insightful inquiry. In response to your suggestion, we have again conducted additional investigations and have provided a more detailed explanation of the database in the revised manuscript.

Answer to the question ‘are these all hospitalization records or both inpatient and outpatient records?’:

To provide further clarity, the CRCCD database predominantly comprises hospitalization records, encompassing a comprehensive array of inpatient medical data. The exclusion of outpatient records from the CRCCD database is attributable to several factors:

(1) **Ambiguous outpatient diagnoses:** In outpatient settings, diagnoses are often preliminary, making it challenging to ascertain the specific condition of an outpatient with certainty.

(2) **Limited clinical testing for outpatients:** Our objective is to gather an extensive range of clinical traits for use as predictors, which includes general information, biochemical tests, biomarkers, medical history, and imaging data, among others. These variables are integrated into our workflow to identify key prognostic markers and uncover new predictors. Importantly, these predictors are not driven by hypotheses and are not preconditioned, thereby better capturing the natural characteristics of the patient and minimizing selection bias.

(3) **complexity and diversity of outpatients:** The sheer number and diversity of outpatients, coupled with the lack of standardized formatting, present significant challenges in data entry and organization, making it difficult to include outpatient records in the CRCCD database. In fact, assuming we have the appropriate research needs, we can still retrieve the patient outpatient records at each hospital sub-center.

Answer to the question ‘Could the authors clarify about the Henan database’:

The Henan Province Clinical Research Center for Cardiovascular Diseases (CRCCD) is strategically located in the North China Plain, encompassing a vast network that spans the densely populated regions of Central China. This feature makes our study cohort relatively representative and avoids small geographic bias to some extent.

To elaborate further, the CRCCD database encompasses data from a network of 11 independent tertiary hospitals, all of which are members of the CRCCD. Every other month, these hospital sub-centers upload the inpatient medical records of patients diagnosed with cardiovascular disease during that period into the CRCCD's centralized database.

11 independent tertiary hospitals subcenters:

- (1) The First Affiliated Hospital of Zhengzhou University Center Hub;
- (2) The First Affiliated Hospital of Henan University of Science and Technology Subcenter;
- (3) Henan Provincial Chest Hospital Subcenter;
- (4) Nanyang Central Hospital Subcenter;
- (5) The First Affiliated Hospital of Xinxiang Medical University Subcenter,
- (6) Xinyang Central Hospital Subcenter;
- (7) Xuchang Central Hospital Subcenter;
- (8) The Second Affiliated Hospital of Zhengzhou University Subcenter;
- (9) Zhengzhou Central Hospital Subcenter;
- (10) The Seventh People's Hospital of Zhengzhou Subcenter;
- (11) Zhoukou Central Hospital Central.

CRCCD used an online electronic data capture system (DCS) designed to ensure accurate data collection. Patient hospitalization records were registered in the DCS by the doctors in charge at each institution. Moreover, patient-identifiable information was dissociated and anonymized. Data were automatically checked for missing or contradictory entries and values out of the normal range. Additional editing and checks for duplicated records were performed by clinical research coordinators at the general office of the registry.

As mentioned in our study, the in-house cohort data was obtained from the electronic health records of the Henan Province Clinical Research Center for Cardiovascular Diseases (CRCCD) in Zhengzhou, Henan, China. Through the online electronic data capture system (DCS) system, we retrospectively enrolled 1256 patients from September 2018 to December 2020 following our inclusion criteria. Of those, 457 patients were excluded due to meeting the exclusion criteria or not signing the consent. Additionally, 87 patients were lost to follow-up during the follow-up period, and 712 patients were finally included and formed an in-house cohort.

In addition, we would like to clarify the BIDMC database in more detail. The external cohort data was accessed from a single-center available database called the Medical

Information Mart for Intensive Care III using the pgAdmin PostgreSQL (version 9.6) Structured Query Language. This database was approved by the Institutional Review Boards of Beth Israel Deaconess Medical Center (BIDMC, Boston, Massachusetts, USA) and the Massachusetts Institute of Technology, included information on 46,520 patients who were admitted to the hospital of BIDMC in Boston, Massachusetts from June 2001 to October 2012. The use of de-identified patient health information is not considered human subjects research, thus eliminating the need for individual patient consent due to the anonymity of the data. We completed the National Institutes of Health Course in the United States, passing the exam of human protection research participants (No.9971167). Moreover, access to the data was approved after completing the Collaborative Institutional Training Initiative (CITI) program “Data or Specimens Only Research” by author Ge Zhang (certification number: 41407001). After this, from the BIDMC center, using the PostgreSQL tool, we identified 1024 eligible patients between June 2001 and October 2012 as an independent external cohort, applying the same inclusion and exclusion criteria as the in-house cohort.

The revised content mentioned above has been incorporated into the 'Data Source' and 'Study Population' subsections of the Methods section (**revised manuscript: line 559-606, page 21-22; line 608-667, page 22-24**), as well as in the 'Population Characteristics Overview' subsection of the Results section (**revised manuscript: line 91-128, page 5-6**).

Thank you again for your kind and constructive comments, and we are sure these will make our study more rigorous, credible and persuasive. We sincerely hope for your understanding and support!

Q2: I have some concerns about the definition of AHF: inoperable severe congenital anomalies or severe valvular anomalies; isolated right heart insufficiency; severe LV structural abnormalities or diastolic dysfunction defined by HFpEF. Can the authors clarify the thresholds for persistence of elevated BNP or NT-proBNP – I assume they were different for HFrEF and HFpEF.

Response:

Thank you for your insightful query. We appreciate the opportunity to clarify the definitions used in our study, and have conducted a thorough investigation in response to your kind suggestion.

Answer to the question '**The definition of AHF**'.

Despite the significant morbidity, it remains challenging to identify patients with advanced HF (AHF). (*JACC Heart Fail. PMID: 32535126*) (*Congest Heart Fail. PMID: 21790965*) This is related in part to lack of a single criterion for AHF. The European Society of Cardiology (ESC) diagnostic criteria for advanced HF is a widely accepted definition, as suggested by most researchers. (*Circulation. PMID: 35363499*), (*Eur J Heart Fail. PMID: 29806100*), (*JACC Heart Fail. PMID: 37589611*)

Therefore, we diligently reviewed two authoritative sources initially: (*2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure, Eur Heart J, PMID: 34447992*) and (*2023 Focused Update of the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure, Eur Heart J, PMID: 37622666*). It is imperative to acknowledge that, since the *2021 ESC Guidelines* were published, several randomized controlled trials have emerged, prompting changes in patient management. The *2023 Focused Update* specifically addresses these changes in treatment recommendations based on new evidence. However, it is important to note that the definitions of heart failure patients and their subtypes have remained unchanged in the *2023 Focused Update*. Therefore, the inclusion and exclusion criteria for our study were primarily derived from the *2021 ESC guidelines*, ensuring consistency and relevance to current clinical practices.

In response to your valuable suggestion, we revisited the definition of AHF as outlined in the *2021 ESC Guidelines*. Specifically, in the '*Section 10 Advanced Heart Failure*', under the '*Subsection 10.1 Epidemiology, Diagnosis, and Prognosis*', the authors present a

comprehensive definition of AHF in (**Table 13**), which has served as a reference point for our study's criteria.

2021 ESC Guidelines Table 13. Criteria for definition of advanced heart failure

All the following criteria must be present despite optimal medical treatment:

1. Severe and persistent symptoms of heart failure [NYHA class III (advanced) or IV].
2. Severe cardiac dysfunction defined by at least one of the following:
 - LVEF $\leq 30\%$
 - Isolated RV failure (e.g., ARVC)
 - Non-operable severe valve abnormalities
 - Non-operable severe congenital abnormalities
 - Persistently high (or increasing) BNP or NT-proBNP values and severe LV diastolic dysfunction or structural abnormalities (according to the definitions of HFpEF).
3. Episodes of pulmonary or systemic congestion requiring high-dose i.v. diuretics (or diuretic combinations) or episodes of low output requiring inotropes or vasoactive drugs or malignant arrhythmias causing >1 unplanned visit or hospitalization in the last 12 months.
4. Severe impairment of exercise capacity with inability to exercise or low 6MWT distance (<300 m) or $pVO_2 < 12$ mL/kg/min or $<50\%$ predicted value, estimated to be of cardiac origin.

[6MWT= 6-minute walk test; ARVC= arrhythmogenic right ventricular cardiomyopathy; BNP= B-type natriuretic peptide; HFpEF= heart failure with preserved ejection fraction; i.v. = intravenous; LV= left ventricular; LVEF= left ventricular ejection fraction; NT-proBNP= N-terminal pro-B-type natriuretic peptide; pVO_2 = peak oxygen consumption; RV= right ventricular]

The definitions employed in our inclusion criteria, such as 'inoperable severe congenital anomalies or severe valvular anomalies,' 'isolated right heart insufficiency,' and 'severe LV structural abnormalities or diastolic dysfunction defined by HFpEF,' are consistent with those outlined in the definitions provided in the **2021 ESC Guidelines**. We apologize for the confusion caused by the previous formatting of this section in our manuscript. To rectify this, we have enriched the manuscript with more detailed information regarding patient inclusion criteria and have improved the formatting in the revised manuscript, specifically in the: 'Study population' subsection of the Methods section (**revised manuscript: line 608-667, page 22-24**).

Answer to the question '**the thresholds for persistence of elevated BNP or NT-proBNP**'.

We acknowledge the oversight regarding the thresholds for persistence of elevated BNP or NT-proBNP in our previous manuscript and apologize for any inconvenience this may have caused. To rectify this, we have incorporated the relevant details into the revised manuscript, providing a clearer understanding of the inclusion criteria for our study population.

'Study population' subsection of the Methods section (**revised manuscript: line 622-627, page 23**):

(v) the persistence of elevated BNP or NT-proBNP values, or severe left ventricular (LV) diastolic dysfunction or structural abnormalities defined by HFpEF (according to the definitions of HFpEF: objective evidence of cardiac structural, functional and serological abnormalities consistent with the presence of left ventricular diastolic dysfunction; NT-proBNP >125 (SR) or >365 (AF) pg/mL; BNP >35 (SR) or >105 (AF) pg/mL; AF= atrial fibrillation, SR= sinus rhythm).

For this aspect, we carefully considered the recommendations provided in the *2021 ESC Guidelines* before establishing our criteria and proceeding with patient inclusion. As outlined in the *2021 ESC Guidelines Table 13*, the definition of AHF includes: Persistently high (or increasing) BNP or NT-proBNP values and severe LV diastolic dysfunction or structural abnormalities (according to the definitions of HFpEF). Consequently, we investigated the definitions of HFpEF as specified in the *2021 ESC Guidelines 'Subsection 8.3 The diagnosis of heart failure with preserved ejection fraction (Page 3630)'*: which necessitates objective evidence of cardiac structural and/or functional abnormalities consistent with the presence of LV diastolic dysfunction/ raised LV filling pressures, including raised NPs (*2021 ESC Guidelines Table 9*). Thresholds for NPs are delineated in *2021 ESC Guidelines Table 9* as follows: (NT-proBNP >125 (SR) or >365 (AF) pg/mL; BNP >35 (SR) or >105 (AF) pg/mL; AF= atrial fibrillation, SR= sinus rhythm).

Answer to the question **'the thresholds for persistence of elevated BNP or NT-proBNP, were they different for HFrEF and HFpEF? '**.

We greatly appreciate your query regarding whether the threshold for elevated BNP or NT-proBNP varies across HF subtypes. Prior to establishing this criterion, we thoroughly reviewed the literature to ensure an informed approach. In fact, the diagnosis of HFmrEF and HFrEF also necessitates the presence of elevated natriuretic peptides (NPs), as evidenced by *2021 ESC Guidelines 'Subsection 5.1 The diagnosis of heart failure with reduced ejection fraction (Page 3619)'*; and *'Subsection 7.1 The diagnosis of heart failure with mildly reduced ejection fraction (Page 3628)'*. The recommended thresholds are BNP > 35 pg/mL or NT-proBNP > 125 pg/mL. Moreover, as indicated by *2021 ESC Guidelines 'Subsection 4.1 Key steps in the diagnosis of chronic heart failure (Page 3614)'* and *'Subsection 4.2 Natriuretic*

peptides (Page 3617)', plasma concentrations of NPs are recommended as initial diagnostic tests in patients with symptoms suggestive of HF to rule out the diagnosis. Elevated concentrations support a diagnosis of HF, are useful for prognostication, and may guide further cardiac investigation. A plasma concentration of B-type natriuretic peptide (BNP) <35 pg/mL, N-terminal pro-B-type natriuretic peptide (NT-proBNP) <125 pg/ mL, or mid-regional pro-atrial natriuretic peptide (MR-proANP) <40 pmol/L make a diagnosis of HF unlikely.

[figure redacted]

Furthermore, the diagnostic algorithm for heart failure, as illustrated in the figure above ([2021 ESC Guidelines Figure1](#)), also demonstrated that the thresholds for elevated BNP or NT-proBNP are consistent across different HF subtypes. These emphasize the importance of these biomarkers as initial diagnostic tests in the assessment of heart failure.

In summary,

the revised criteria content can be found in the 'Study Population' subsection of the Methods section (**revised manuscript: line 607-666, page 22-24**):

Study population

The inclusion criteria were as follows:

- (1) Over 18 years old.
- (2) Diagnosis with AHF using the International Classification of Disease 9th revision (ICD9) after the first admission, referring to the 2021 ESC advanced HF diagnostic criteria.⁸

At least two of the following three criteria must be met despite the treatment:

(a) Severe and persistent HF symptoms [New York Heart Association (NYHA) class III or IV].

(b) Severe cardiac dysfunction was defined by at least one of the following:

- (i) ejection fraction less than or equal to 30%;

(ii) isolated right ventricular (RV) failure (e.g., arrhythmogenic right ventricular cardiomyopathy (ARVC));

(iii) inoperable severe valvular anomalies;

(iv) inoperable severe congenital anomalies;

(v) the persistence of elevated BNP or NT-proBNP values, or severe left ventricular (LV) diastolic dysfunction or structural abnormalities defined by HFpEF (according to the definitions of HFpEF: objective evidence of cardiac structural, functional and serological abnormalities consistent with the presence of left ventricular diastolic dysfunction; NT-proBNP >125 (SR) or >365 (AF) pg/mL; BNP >35 (SR) or >105 (AF) pg/mL; AF= atrial fibrillation, SR= sinus rhythm).

(c) Low output paroxysmal HF requiring positive inotropes and vasoactive drugs; or pulmonary or systemic congestion episodes requiring high intravenous dose diuretics; or hospitalization in the past 12 months; or malignant arrhythmia resulting in >1 unscheduled visit.

Episodes of pulmonary or systemic congestion requiring high-dose i.v. diuretics (or diuretic combinations) or episodes of low output requiring inotropes or vasoactive drugs or malignant arrhythmias causing >1 unplanned visit or hospitalization in the last 12 months.

(d) Severe impairment of exercise capacity with inability to exercise or low 6-minute walk test (6MWT) distance (<300 m) or pVO₂ <12 mL/kg/min or <50% predicted value, estimated to be of cardiac origin.

Detailed information on AHF operationalized identification in the study was provided in **(Revised Supplementary Table 18)**.

(3) Diagnosis of renal insufficiency, characterized by an estimated glomerular filtration rate (eGFR) of less than 90 ml/(min·1.73m²) and abnormalities in at least one of the following: blood creatinine or blood urea nitrogen levels.

Exclusion criteria:

(1) Initial onset of acute heart failure.

(2) Diagnosis of primary renal disease.

(3) Diagnosis of an infectious disease or malignant tumor.

(4) Receiving renal dialysis treatment.

(5) Incomplete clinical case data: with missing >20% of individual data on relevant covariates necessary for research.

(6) Hospitalization time <2 days.

In fact, the recruitment of patients to form a cohort involves much considerations. Therefore, we have re-provided an additional operationalized definition of AHF, which can be found in the **Revised Supplementary Table 18** in the **Revised Supplemental Material**. We hope that these materials will provide greater clarity on the definition of AHF in clinical practice, rendering some padding and assistance for future research!

Supplementary Table 18. Advanced Heart Failure Operationalized Definition

Criteria	Description	Method	Definition
1. Episodes of congestion, low output, or malignant arrhythmias	hospitalization or ED visit for HF or malignant arrhythmia *Also included hospitalization for LVAD, heart transplant	Code based	HF hospitalization: Primary ICD-9 codes 428.X, 402.X1, 404.X1, 404.X3 or ICD-10 codes I11.0, I13.0, I13.2, or I50 VT/ VF hospitalization or ED visit ^a : Primary ICD9 427.1, 427.4, 427.4X, 427.5 or ICD10 I47.2, I49.01, I49.02, I46.9 LVAD is ICD9 procedure 37.66 or ICD 10-PCS 02HA0QZ or ICD10 PCS 02HA3QZ or ICD10 PCS 02HA4QZ Heart transplant is ICD9 procedure 37.51 ICD 10 PCS 02YA0Z0 HF ED visit ^b : same codes as hospitalization but also include codes where HF is in 2 nd or 3 rd position if first position is • ICD9: 276.6, 514, 518.4, 518.81, 518.83, 518.84, 786, 782.3 • ICD10: E8770, E8779, J182, J811, J810, J9600, J9690, J9610, J9620, R069, R0601, R0602, R0600, R0609, R600, R601, R609 Each event is a potential index date; rest of criteria need to be met within 12 months
2. Severe cardiac dysfunction (any of):			
	EF≤30%	Echo	EF≤30%
	Isolated RV failure	Echo	Moderate, moderate severe, severe RV dysfunction
	Nonoperable severe valve	Echo/ manual review	Severe valvular stenosis or regurgitation with manual review to confirm patient not considered to be surgical candidate by physician
	Nonoperable severe congenital anomalies	Echo/ manual review ^c	Manual review of those w/ a congenital echo Billing codes were not sensitive or specific
	Severe diastolic dysfunction or LV structural abnormalities and HFrEF or HFmrEF	Echo	Echo (all of the following): • EF≥40% • LV hypertrophy (LV mass index≥115 men or ≥95 women) or (LV mass index missing and increased LV wall thickness as a physician final impression) OR LA enlargement: LAVI>34 or (LAVI missing and moderately/moderate-severely/severely enlarged LA) • Severe diastolic dysfunction (grade 2, 3 or 4)^a defined as any of the following: - o Stated grade 2/3/4 diastolic dysfunction in final impressions - o If no diastolic function grade given, if mitral valve E/A present then severe diastolic dysfunction if: ▪ E/A≥0.8 + either LAE or E/e'≥10 (=grade 2) ▪ E/A>2 + either LAE or E/e'≥10 (=grade 3/4)

			- o If above missing, then is severe diastolic function if either: ▪ "LV filling pressure increased" listed in final impressions ▪ E/e² ≥ 10 Elevation in natriuretic peptides was not required due to inconsistent use in clinical practice. It has also been demonstrated that elevated natriuretic peptide levels are not required for the diagnosis of HFpEF^f.
3. Severe impairment of exercise (any of):			
	Low peak V _{O₂}	Manual review ^d	≤12 mL/kg/min or ≤50% predicted ^g
	Short 6MWT distance	Manual review ^d	≤300m
	Inability to exercise due to HF	Manual review ^c	Inability to walk/ a city block/ anything that describes significant exercise mobility limitations due to HF (dyspnea, weakness, lightheadedness)
4. Severe and persistent symptoms of HF	NYHA class III or IV	Manual chart review ^c	Symptoms of HF at rest or with minimal exertion
5. Low output paroxysmal HF		Manual chart review ^c	Low output paroxysmal HF requiring positive inotropes and vasoactive drugs, or pulmonary or systemic congestion episodes requiring high intravenous dose diuretics
6. Despite GDMT		Manual chart review ^c	All of the above #1-5 despite attempts at optimal therapy

6MWT= six minute walk test, ED= emergency department, EF= ejection fraction, GDMT= guideline directed medical therapy, HF= heart failure, LA= left atrial, LAVI=left atrial volume index, LV= left ventricular, LVAD= left ventricular assist device, RV=right ventricular, VT/VF= ventricular tachycardia/ ventricular fibrillation

Criteria are operationalized based on 2021 ESC Advanced Heart Failure Criteria in Eur J Heart Fail 2021, 42:3599-3726

^aMcDonald KM, et al. Am Heart J 2002; 144(3):413-21

^bBlecker S, et al. Am Heart J 2014; 168(6):901-08

^cManual review by heart failure cardiologist

^dManual review by trained research coordinator

^eNagueh SF, et al. J Am Soc Echocardiogr 2016; 29(4):277-314

^fAnjan VY, et al. Am J Cardiol 2012; 110:870-76

^gMehra MR, et al. J Heart Lung Transplant 2016; 35(1):1-23

Once again, thanks for your constructive feedback, and we are sure that your suggestions have enriched and refined our research. We hope that these revisions meet your expectations and address your concerns.

Q3: The Henan cohort seems to be predominantly HFpEF with a surprisingly lower (<20%) HFrEF. Would it be better to focus on a specific type of heart failure since AHF may be different in terms of prognosis and treatment options in HFpEF and HFrEF?

Response:

Thank you very much for your professional comments. We appreciate the opportunity to clarify the cohort baseline characteristics.

Answer to the question ' The Henan cohort seems to be predominantly HFpEF with a surprisingly lower (<20%) HFrEF'.

To begin with, we would like to clarify our findings:

- (1) In the **CRCCD cohort (China populations)**: HF subtypes were distributed as 58.2% HFpEF, 23.5% HFmrEF, and 18.3% HFrEF (**Revised Supplementary Table 1**);
- (2) In the **BIDMC cohort (U.S. populations)**: HF subtypes were distributed as 66.9% HFpEF, 11.5% HFmrEF, and 21.6% HFrEF (**Revised Supplementary Table 2**).

As you kindly highlighted, it's an interesting observation that despite significant heterogeneity in the populations (especially in terms of geography and ethnicity), the proportions of HF subtypes are remarkably similar in both the Chinese and U.S. cohorts, with HFpEF constituting a large portion of the population, while HFrEF and HFmrEF are less prevalent. This caused us to think about potential causes. We would like to discuss the following aspects:

(1) From the perspective of pathogenesis:

Renal dysfunction (RD) can worsen cardiovascular function causing hypertension and vascular calcification. In turn, HF may worsen renal function, through the effects of neurohormonal and inflammatory activation, increased venous pressure and hypoperfusion (*Eur J Heart Fail*, PMID:26861140; *Eur J Heart Fail*, PMID:30338885). Notably, CKD and worsening renal function both appear more common in HFpEF as compared to in HFmrEF and HFrEF, perhaps due to shared pathophysiological mechanisms, as evidenced by **2021 ESC Guidelines**, ([Subsection 13.6 Kidney dysfunction \(Page 3669\)](#)).

(2) From the perspective of epidemiology:

HFpEF differs from HFrEF and HFmrEF in that HFpEF patients could be older. CKD and atrial fibrillation comorbidities are found to be more common in patients with HFpEF than in those with HFrEF or HFmrEF (*Nat Rev Cardiol*, PMID: 32231333) (**2021 ESC Guidelines**, [Subsection 8.2 Clinical characteristics of patients with heart failure with preserved ejection fraction \(page 3630\)](#)). In a large, multi-center, prospective study conducted by the Chronic Renal Insufficiency Cohort (CRIC) in the U.S., researchers observed that the rate of first hospitalization during follow-up was higher for HFpEF than that of HFrEF, suggesting that while HFpEF may be more prevalent in CKD. However, the rates of hospitalizations were similar between HFrEF and HFmrEF (*J Am Coll Cardiol*, PMID: 31146814). Furthermore, another study's findings also highlighted that in a contemporary cohort of hospitalized HF, comorbid kidney dysfunction appears to be highly prevalent, particularly for HFpEF, and has been stable over time (*J Am Coll Cardiol*, PMID: 33989713). Another study also reported that The prevalence of HFpEF is higher in Internal Medicine units, often associated with a number of comorbidities, including impaired renal function, leading to less favorable prognosis (*Eur J Intern Med*, PMID: 238308763).

Given that our cohort specifically consisted of patients with advanced heart failure and renal insufficiency, such distribution pattern in our study exactly corroborates previous reports. It is noteworthy that most of these research findings are based on Western populations. In addition to observing consistent results in our BIDMC US cohort, we also found a similar distribution pattern in our CRCCD Chinese cohort. This further supports the notion that the deterioration of the cardio-renal axis is more likely to occur in the HFpEF subtype.

Answer to the question 'Would it be better to focus on a specific type of heart failure since AHF may be different in terms of prognosis and treatment options in HFpEF and HFrEF'.

Thank you for your expert insights, and we will address the following aspects in our discussion:

(1) Why we should be concerned about advanced heart failure combined with renal dysfunction:

Many patients with HF progress into a phase of advanced HF, characterized by persistent symptoms despite maximal therapy. (*J Card Fail*, PMID:25953697; *JACC Heart Fail*, PMID: 32535126) The prevalence of advanced HF is increasing due to the growing number of patients with HF, ageing of the population, and better treatment and survival of HF. Prognosis remains poor, with a 1-year mortality ranging from 25% to 75% (*N Engl J Med*, PMID: 11794191; *N JACC Heart Fail*, PMID: 27395350).

CKD has consistently been recognized as a prevalent comorbidity in HF, (*Eur J Heart Fail*, PMID: 28371075; *Eur Heart J*, PMID: 37622666) and when present, carried the highest population attributable risk for all-cause mortality (ACM) and AHF hospitalization among all

comorbidities. (*Circulation*, PMID: 35226558; *Nat Rev Nephrol*, PMID: 27573728; *Int J Cardiol*, PMID: 28528982; *JACC Heart Fail*, PMID: 30606484). The complex interaction between HF and renal dysfunction accelerated disease progression, driven by neurohormonal and inflammatory activation, elevated venous pressure, and hypoperfusion (*Eur Heart J*, PMID: 34447992). Clinically, AHF patients with RD may encounter additional harm because such patients often received lower doses of drugs, and diagnostic tests using contrast media are avoided (*Eur J Heart Fail*, PMID: 26635076). While emerging biomarkers like natriuretic peptides provided some post-discharge prognostic value, they are not specific enough for AHF or RD cases, constraining their clinical utility (*JACC Heart Fail*, PMID: 36114137; *Ann Intern Med*, PMID: 31842204; *JAMA Cardiol*, PMID: 30140899; *J Card Fail*, PMID: 30965102; *J Pers Med*, PMID: 35743683). The efficacy of management and prognostication guidance using serial natriuretic peptide measurements remained unestablished (*Eur J Heart Fail*, PMID: 37098791). Additionally, heterogeneity within HF and RD populations reduced the efficacy of a single biomarker (*Circulation*, PMID: 29792299). Therefore, comprehensive survival predictions should be prioritized in this high-risk population for clear prognostication and targeted interventions (*Circ Heart Fail*, PMID: 28765150; *J Am Coll Cardiol*, PMID: 33989713; *Eur Heart J*, PMID: 36691956).

(2) Why we didn't develop a modeling system for just a single subtype:

(I) Initially, there is a substantial overlap of clinical characteristics, risk factors, patterns of cardiac remodelling, and outcomes among the LVEF categories in HF (*2021 ESC Guidelines*, 'Subsection 7.2 Clinical characteristics of patients with heart failure with mildly reduced ejection fraction (page 3629)').

(II) Moreover, despite that HFpEF is generally considered to confer a better survival than HFrEF or HFmrEF, but most observational studies show that this difference is negligible. (*2021 ESC Guidelines*, 'Subsection 3.3.3 Natural history and prognosis (page 3614)'; *Temporal Trends in the Incidence of and Mortality Associated With Heart Failure With Preserved and Reduced Ejection Fraction*, *JACC Heart Fail*, PMID: 30007560; *A contemporary appraisal of the heart failure epidemic in Olmsted County, Minnesota, 2000 to 2010. JAMA Intern Med*. PMID: 25895156).

(III) In a study involving patients from multiple healthcare systems within the Cardiovascular Research Network (CVRN), it was found that the impact of renal function on the risk of hospitalization or mortality was similar for both HFpEF and HFrEF patients (*J Am Coll Cardiol*, PMID:31146814; *Circ Cardiovasc Qual Outcomes*, PMID:23685625).

(IV) Although much of the evidence suggested that HFpEF could be the predominant subtype in populations with AHF combined with RD, it is important to acknowledge that other subtypes of HF also evolve into advanced stages or be associated with renal complications (2021 ESC Guidelines, 'Subsection 10.1 Epidemiology, diagnosis, and prognosis'). In addition, transition in ejection fraction over time is common in the same heart failure patient, indicating that the HF subtype does shift in response to evolving clinical conditions (*Dynamic Trajectories of Left Ventricular Ejection Fraction in Heart Failure, J Am Coll Cardiol, PMID: 30071987, 2021 ESC Guidelines, 'Subsection 3.3.3 Natural history and prognosis*).

(V) In fact, RD and loop diuretic resistance often characterize the clinical course of patients with AHF. (*The use of diuretics in heart failure with congestion - a position statement from the Heart Failure Association of the European Society of Cardiology. Eur J Heart Fail. PMID: 30600580, 2021 ESC Guidelines, '10. Advanced heart failure, Subsection 10.2.1 Pharmacological therapy and renal replacement*). Importantly, there is little direct evidence to support any recommendations for treatment of AHF patients with CKD (2021 ESC Guidelines, 'Subsection 13.6 Kidney dysfunction').

(3) From Specificity to Universality: Establish a universal prognostic tool:

Our aim is to develop a clinically practical tool that is not only precise in performance but also possesses universality and generalizability, making it applicable to the majority of heart failure patients. From the perspective of artificial intelligence model training, if we were to train and develop the model system specifically for one subtype (e.g., HFpEF), the model might become specific only to HFpEF. However, clinical encounters often involve patients from all three subtypes, and transition in ejection fraction over time is common in the same heart failure patient. Therefore, if we face a patient who is not classified as HFpEF, the question arises whether we should use this system for prediction, and whether the prediction results would be credible and accurate enough to be considered for clinical decision-making.

In summary, including all subtypes of HF in our cohort is driven by the need to develop a tool that is applicable to the broader population of patients with AHF and RD, rather than limiting its utility to a specific subtype. Thus, while the focus on a specific subtype might be beneficial in some contexts, the inclusive approach taken in our study could be justified by the reasons outlined above.

In response to your valuable suggestions, we have further elaborated on this section, as detailed in the 'Introduction' section (**revised manuscript: lines 33-88, pages 3-4**) and the 'Population Characteristics Overview' subsection of the Results section (**revised manuscript: lines 92-128, pages 5-6`**).

We sincerely appreciate your constructive feedback, which has undoubtedly enhanced and sharpened our research. We trust that these revisions align with your expectations and adequately address your concerns.

Q4: Missing values – could the authors state how much data was missing?

Response:

We appreciate your insightful comments on the issue of missing data. In our initial manuscript, we briefly outlined our approach to handling missing values but failed to provide a detailed description. We apologize for any confusion this may have caused. In the revised manuscript, we have included a more detailed account of the extent of missing data and the methods employed to address it.

(1) Data collection:

We retrieved the patient data within the initial 24 hours following admission. 93 easily accessible variables were listed as candidates for constructing the survival assessment system subsequently. Further details can be found in the 'Data collection ' subsection of the Methods section (**revised manuscript: lines 669-710, pages 24-26**).

(2) Missing data handling:

The (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis (TRIPOD) guideline suggested applying multiple imputations when missing data were present, as complete case analyses can lead to reduced sample size, biased estimates, and loss of Information (*Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD): explanation and elaboration. Ann Intern Med. PMID: 25560730; Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. BMJ. PMID: 19564179*). The BIDMC cohort presented no instances of missing data, while the extent of missingness in the CRCCD cohort is detailed in (revised Supplementary Table 19).

Supplementary Table 19. Missingness report for Henan Province Clinical Research Center for Cardiovascular Diseases Cohort.

	Missing number	Total number	Missing percentage	Variable type
NYHA	93	712	13.06%	Categorical
CKD stage	27	712	3.79%	Categorical
Fg	1	712	0.14%	continuous
BNP	1	712	0.14%	continuous
CRP	4	712	0.56%	continuous
GLU	2	712	0.28%	continuous
RVID	9	712	1.26%	continuous
IVS	10	712	1.40%	continuous
LVID	8	712	1.12%	continuous
LVPWT	8	712	1.12%	continuous
AOAD	9	712	1.26%	continuous
LAD	8	712	1.12%	continuous
PVAD	9	712	1.26%	continuous
RALRD	43	712	6.04%	continuous
RASID	54	712	7.58%	continuous
MVE	72	712	10.11%	continuous
MVA	82	712	11.52%	continuous
EF	13	712	1.83%	continuous
SV	44	712	6.18%	continuous
FS	44	712	6.18%	continuous

Abbreviations:

NYHA, New York Heart Association Classification System; CKD, chronic kidney disease; CRP, C-reactive protein; GLU, glucose; RVID, right ventricle internal diameter; IVS, interventricular

septal thickness; LVID, left ventricle internal diameter; LVPWT, left ventricular posterior wall thickness; AOAD, aortic valve annulus diameter; LAD, left atrium diameter; PAVD, pulmonary artery valve diameter; RALRD, right atrium lateral diameter; RASID, right atrium supero-inferior diameter; MVE, early diastolic mitral wave velocity; MVA, late diastolic mitral wave velocity ratio; EF, ejection fraction; SV, stroke volume; FS, left ventricular fraction shortening

Initially, our data was presumed to be Missing At Random (MAR), aligning with the assumptions of prior studies (*Eur J Heart Fail. PMID: 30919549; JAMA Netw Open. PMID: 34767024*). We conducted Little's test to confirm that the data were not missing completely at random (MCAR) (*A Test of Missing Completely at Random for Multivariate Data with Missing Values, Journal of the American Statistical Association, doi:10.1080/01621459.1988.10478722*). The result was depicted in the **(revised Supplementary Table 19)**. Therefore, multiple imputation was performed through the method of chained equations procedure, ensuring a statistically robust approach to addressing missing data in our study (*Multiple imputation by chained equations: what is it and how does it work? Int J Methods Psychiatr Res. PMID: 21499542*)

Supplementary Table 19. Use Little's test to assess for missing completely at random (MCAR), suggesting our data is unlikely to be MCAR

Little's test	Chi.square	(Degrees of freedom used for chi-square)	P-value
value	1483.688	1104	1.26e-13

Little, R. J. A. (1988). A test of missing completely at random for multivariate data with missing values. *Journal of the American Statistical Association*, 83(404), 1198–1202.

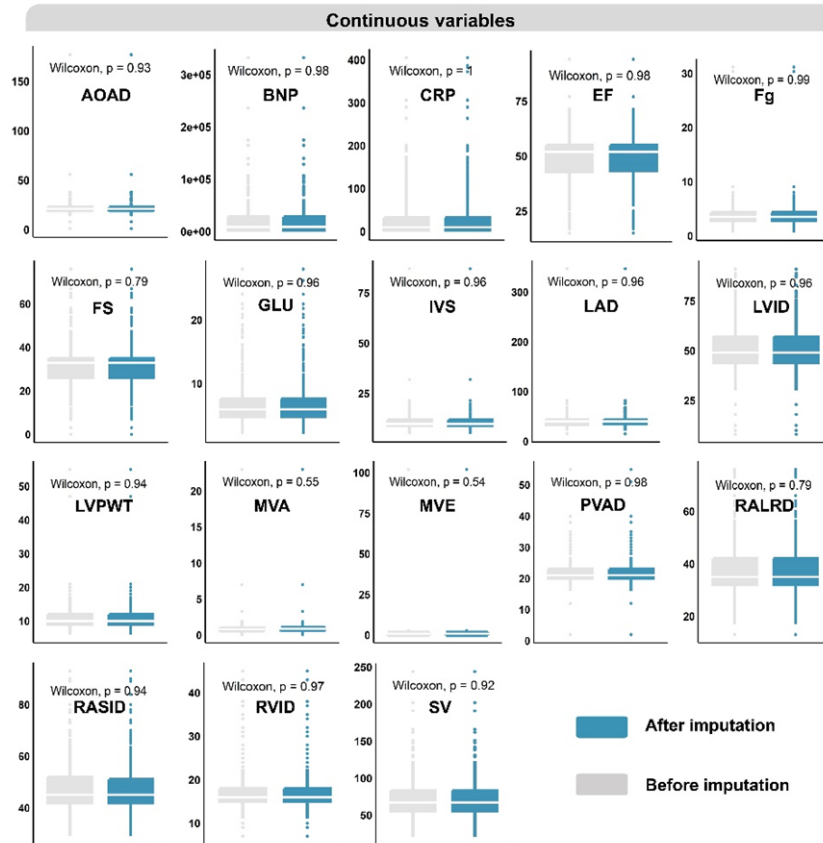
Multiple imputation with chained equations (MICE) specified a multivariate model through conditional distributions, where each variable with missing data is imputed conditionally on all other variables. In our dataset with p (p=93) variables and k (k=20) variables missing data (with p-k complete cases), the procedure was as follows:

- (1) For each of the k variables with missing data, an imputation model was specified.
 - (a) numeric data: bayesian linear regression imputation.
 - (b) factor data with 2 levels: logistic regression imputation.
 - (c) factor data with >2 unordered levels: polytomous regression imputation.

- (d) factor data with >2 ordered levels: proportional odds model.
- (2) Imputation values are drawn from the overall distribution of observed values for each variable with missing data. These initial imputed values are refined in subsequent steps.
- (3) For the first variable with missing data:
- (a) Regression is performed based on other variables (including the complete data of the first variable; and other observed or imputed variables).
- (b) Regression parameters: extract estimated regression coefficients and the variance-covariance matrix of the regression model from the previous regression (and for linear regression models fitted for continuous variables, the estimated variance of the residuals).
- (c) Parameters are randomly perturbed to reflect the uncertainty in data generation. The conditional distribution for each sample with missing data in the first variable is determined using the perturbed regression coefficients.
- (d) Imputation values are then selected for each missing datum based on the conditional distribution.
- (4) Steps 3 and 4 are repeated in a cycle, imputing each variable with missing data.
- (5) The process of steps 3 and 4 is repeated 20 iterations. The final imputed values form the first completed imputed dataset.
- (6) Steps 3 to 5 are repeated 5 times to generate 5 imputed datasets.
- (7) Construction of the multiple imputed datasets is completed and now M datasets are obtained (assuming M=5). These datasets are then treated as complete for statistical analysis. The results were pooled into one dataset by applying Rubin's rules.

(3) Reliability of Missing Data Handling

In addressing missing data for continuous and categorical variables, our analysis revealed a uniform pattern in the data distribution before and after interpolation, as detailed in **(revised Supplementary Fig. 7)** and **(revised Supplementary Fig. 8)**. This consistency was validated using the Wilcoxon rank-sum test for continuous variables and the chi-squared test for categorical variables, ensuring the integrity and reliability of our data analysis.



Supplementary Figure 7. In the process of managing missing continuous variable data within the in-house cohort (Henan Provincial Cardiovascular Disease Clinical Research Center), we observed a consistent data distribution pattern both pre- and post-interpolation. The comparison was assessed using the Wilcoxon rank-sum test. CRP, C-reactive protein; GLU, glucose; RVID, right ventricle internal diameter; IVS, interventricular septal thickness; LVID, left ventricle internal diameter; LVPWT, left ventricular posterior wall thickness; AOAD, aortic valve annulus diameter; LAD, left atrium diameter; PAVD, pulmonary artery valve diameter; RALRD, right atrium lateral diameter; RASID, right atrium supero-inferior diameter; MVE, early diastolic mitral wave velocity; MVA, late diastolic mitral wave velocity ratio; EF, ejection fraction; SV, stroke volume; FS, left ventricular fraction shortening.

Categorical variables					
	After imputation	Before imputation	Pearson's Chi-squared test X-squared	Pearson's Chi-squared test df	Pearson's Chi-squared test P-value
Cohort size	N = 712	N = 712			
NYHA (%)	n = 712 (100.00(%))	n = 619 (86.94(%))	2.1286	3.0	0.5461
IV	26 (3.65%)	16 (2.58%)			
III	86 (12.08%)	67 (10.82%)			
II	225 (31.6%)	193 (31.18%)			
I	375 (52.67%)	343 (55.41%)			
CKD stage (%)	n = 712 (100.00(%))	n = 685 (96.21(%))	2.3321	5.0	0.8015
V	168 (23.6%)	168 (24.53%)			
IV	136 (19.1%)	135 (19.71%)			
IIIb	128 (17.98%)	127 (18.54%)			
IIIa	22 (3.09%)	13 (1.9%)			
II	249 (34.97%)	233 (34.01%)			
I	9 (1.26%)	9 (1.31%)			

Supplementary Figure 8. In the process of managing missing categorical variable data within the Henan Provincial Cardiovascular Disease Clinical Research Center cohort, we observed a consistent data distribution pattern both pre- and post-interpolation. The comparison was assessed using the chi-squared contingency table tests. NYHA, New York Heart Association Classification System; CKD, chronic kidney disease.

Following your insightful recommendations, we have expanded upon this aspect in the 'Missing data handling' subsection of the Methods section (**revised manuscript: lines 712-764, pages 27-28**).

We are grateful for your constructive feedback, which has significantly refined and enriched our study. We hope that these revisions meet your expectations and effectively address your concerns.

Reviewer #2 (Remarks to the Author):

The submitted manuscript discusses an important issue: predicting advanced heart failure in patients with renal dysfunction. The authors have developed and implemented an AI-based tool that offers improved accuracy and accessibility compared to the available risk scores. However, there are some issues that should be clarified in order to fully appreciate the comprehensive methodology the authors have used:

Response:

Thanks for your insightful advice. We have carefully rechecked and revised our manuscripts again. As you mentioned later, our manuscript has some pending issues that need to be further clarified, such as checklist, clarification of the study outcome, details of the consensus assessment strategy, discussion of the missing data as well as the external validation.

In response to your valuable comments, the revised manuscript was now modified and restructured for proper to improve readability and comprehension. We have included additional data analyses and added detailed discussion to properly interpret the prediction system AIHFLevel and to give it fuller clinical implications. Detailed information and raw data related to the analysis results have been added to enhance the reproducibility of the research data, and are now included in the **revised Supplementary Tables 1-20**. Every single figure and Supplementary figure have been polished and restructured for clarity and readability in the **revised Figures 1-7** and **revised Supplementary Figures 1-10**. Please find **the overview of the revised manuscript and research** in the responses to Reviewer #1 in **(Remarks to the Author)**. We will clarify and address each of your suggestions in turn.

We earnestly hope that our revisions will gain your esteemed approval!

Q1: As the authors have developed a prediction system, it would be beneficial if they could provide the TRIPOD statement and checklist.

Response:

We greatly value your expert feedback and appreciate the chance to provide further clarification on our study through the re-provided checklist.

As you rightly pointed out, studies developing or validating a multivariable prediction model share specific challenges for researchers (*PLoS Med*, PMID: 23393430). Several reviews have evaluated the quality of published reports that describe the development or validation prediction models (*PLoS Med*, PMID: 22629234). For example, Mallett and colleagues examined 47 reports presenting new prediction models in cancer. Reporting was found to be poor, with insufficient information described in all aspects of model development, from descriptions of patient data to statistical modelling methods (*BMC Med*, PMID: 20353578). Collins and colleagues evaluated the methodological conduct and reporting of 39 reports describing the development of models. Reporting was also found to be generally poor, with key details on which predictors were examined, the handling and reporting of missing data, and model-building strategy often poorly described (*BMC Med*, PMID: 21902820). Bouwmeester and colleagues evaluated 71 reports, published in 2008 in six high-impact general medical journals, and likewise observed an overwhelmingly poor level of reporting (*PLoS Med*, PMID: 22629234). These and other reviews provide a clear picture that, across different disease areas and different journals, there is a generally poor level of reporting of prediction model studies. Furthermore, these reviews have shown that serious deficiencies in the statistical methods, use of small data sets, inappropriate handling of missing data, and lack of validation are common (*Circulation*, PMID: 20679549). Such deficiencies ultimately lead to prediction models that are not or should not be used. It is therefore not surprising, and fortunate, that very few prediction models, relative to the large number of models published, are widely implemented or used in clinical practice.

TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis) Statement serves as a guideline specifically designed for the reporting of studies developing or validating a multivariable prediction model, whether for diagnostic or prognostic purposes. The TRIPOD Statement is a checklist of 22 items that are essential for good reporting of studies developing or validating multivariable prediction models.

We downloaded the TRIPOD checklist from www.tripod-statement.org, and reported our study in line with the TRIPOD recommendations (**Revised Supplementary Table 17**) (*Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD): explanation and elaboration. Ann Intern Med. PMID: 25560730*).

Supplementary Table 17. TRIPOD Checklist: Prediction Model Development and Validation

Section/Topic	Item	Checklist Item		Page
Title & abstract				
Title	1	D;V	Identify the study as developing and/or validating a multivariable prediction model, the target population, and the outcome to be predicted.	Title (1)
Abstract	2	D;V	Provide a summary of objectives, study design, setting, participants, sample size, predictors, outcome, statistical analysis, results, and conclusions.	Abstract (2)
Introduction				
Background and objectives	3a	D;V	Explain the medical context (including whether diagnostic or prognostic) and rationale for developing or validating the multivariable prediction model, including references to existing models.	Introduction, para 1-3 (3)
	3b	D;V	Specify the objectives, including whether the study describes the development or validation of the model or both.	Introduction, para 4 (4)
Methods				
Source of data	4a	D;V	Describe the study design or source of data (e.g., randomized trial, cohort, or registry data), separately for the development and validation data sets, if applicable.	Methods, para 1, 2 and 6 (14-17; 20; 22)
	4b	D;V	Specify the key study dates, including start of accrual; end of accrual; and, if applicable, end of follow-up.	Methods, para 1, 2 (14-15; 17)
Participants	5a	D;V	Specify key elements of the study setting (e.g., primary care, secondary care, general population) including number and location of centres.	Methods, para 1 (14-15)
	5b	D;V	Describe eligibility criteria for participants.	Methods, para 2 (16-17)
	5c	D;V	Give details of treatments received, if relevant.	NA
Outcome	6a	D;V	Clearly define the outcome that is predicted by the prediction model, including how and when assessed.	Methods, para 5 (19)
	6b	D;V	Report any actions to blind assessment of the outcome to be predicted.	NA
Predictors	7a	D;V	Clearly define all predictors used in developing or validating the multivariable prediction model, including how and when they were measured.	Methods, para 3 (17-18)
	7b	D;V	Report any actions to blind assessment of predictors for the outcome and other predictors.	NA
Sample size	8	D;V	Explain how the study size was arrived at.	Methods, para 2 (16)
Missing data	9	D;V	Describe how missing data were handled (e.g., complete-case analysis, single imputation, multiple imputation) with details of any imputation method.	Methods, para 4 (18-19)
Statistical analysis methods	10a	D	Describe how predictors were handled in the analyses.	Methods, para 3 and 6 (17-18; 20)
	10b	D	Specify type of model, all model-building procedures (including any predictor selection), and method for internal validation.	Methods, para 6 (20)
	10c	V	For validation, describe how the predictions were calculated.	Methods, para 6 (20-21)
	10d	D;V	Specify all measures used to assess model performance and, if relevant, to compare multiple models.	Methods, para 6 (21)

	10e	V	Describe any model updating (e.g., recalibration) arising from the validation, if done.	NA
Risk groups	11	D;V	Provide details on how risk groups were created, if done.	Methods, para 7 (22-23)
Development vs. validation	12	V	For validation, identify any differences from the development data in setting, eligibility criteria, outcome, and predictors.	Methods, para 6 (21-22); Results, para 8 (9)
Results				
Participants	13a	D;V	Describe the flow of participants through the study, including the number of participants with and without the outcome and, if applicable, a summary of the follow-up time.	Results, para 1 (4-5)
	13b	D;V	Describe the characteristics of the participants (basic demographics, clinical features, available predictors), including the number of participants with missing data for predictors and outcome.	Methods, para 4 (18-19)
	13c	V	For validation, show a comparison with the development data of the distribution of important variables (demographics, predictors and outcome).	Methods, para 6 (22); Results, para 8 (9-10)
Model development	14a	D	Specify the number of participants and outcome events in each analysis.	Results, para 1 (4-5)
	14b	D	If done, report the unadjusted association between each candidate predictor and outcome.	Results, para 2 (5)
Model specification	15a	D	Present the full prediction model to allow predictions for individuals (i.e., all regression coefficients, and model intercept or baseline survival at a given time point).	Results, para 3 (5-6)
	15b	D	Explain how to use the prediction model.	Results, para 9 (10-11)
Model performance	16	D;V	Report performance measures (with CIs) for the prediction model.	Results, para 3, 4, 5, 6, and 7 (5-9)
Model-updating	17	V	If done, report the results from any model updating (i.e., model specification, model performance).	Results, para 3, 4, 5, 6, and 7 (5-9)
Discussion				
Limitations	18	D;V	Discuss any limitations of the study (such as nonrepresentative sample, few events per predictor, missing data).	Discussion, para 6 (14)
Interpretation	19a	V	For validation, discuss the results with reference to performance in the development data, and any other validation data.	Discussion, para 3 (12)
	19b	D;V	Give an overall interpretation of the results, considering objectives, limitations, results from similar studies, and other relevant evidence.	Discussion, para 1 (11)
Implications	20	D;V	Discuss the potential clinical use of the model and implications for future research.	Discussion, para 5, and 7 (13-14)
Other information				

Supplementary information	21	D;V	Provide information about the availability of supplementary resources, such as study protocol, Web calculator, and data sets.	Discussion, para 5 (13)
Funding	22	D;V	Give the source of funding and the role of the funders for the present study.	Acknowledgement

*Items relevant only to the development of a prediction model are denoted by D, items relating solely to a validation of a prediction model are denoted by V, and items relating to both are denoted D;V.

The revised content mentioned above has been incorporated into the 'Data Source' subsection of the Methods section (**revised manuscript: line 559-606, page 21-22**), as well as in the (**Revised Supplementary Table 17**).

We are deeply grateful for your constructive advice, which we believe will significantly enhance the rigor, credibility, and persuasiveness of our study. We sincerely hope for your understanding and approval.

Q2: When defining a study outcome, it would be helpful if the authors could provide more details. Why have they used all-cause mortality and not a specific cause, at least for a sensitivity analysis? What particular length of survival are the authors looking for? Later on, they report survival at 6/12/24/30 months, and presumably, that's what should be specified

Response:

We greatly appreciate your kind and insightful inquiry. In response, we have undertaken additional data analyses and expanded our discussion of the study outcomes in the revised manuscript for a more comprehensive understanding.

The first aspect that requires further clarification is the methodology for collecting follow-up information and identifying study outcomes. For the CRCCD cohort, this process involved extracting case system records, contacting patients and their relatives, and consulting with referring physicians via telephone or mail. In contrast, for the BIDMC cohort, we gathered information from the publicly available single-center database, Medical Information Mart for Intensive Care III, utilizing the pgAdmin PostgreSQL (version 9.6) Structured Query Language (**revised manuscript: line 767-780, page 28**).

(1) Why we choose the all-cause mortality (ACM) as the primary endpoint?

(i) The Critical Role of ACM:

More and more patients with HF progress into a phase of advanced HF (AHF), characterized by persistent symptoms despite maximal therapy. (*J Card Fail*, PMID:25953697; *JACC Heart Fail*, PMID: 32535126) The prevalence of AHF is increasing due to the growing number of patients with HF, ageing of the population, and better treatment and survival of HF. Prognosis remains poor, with a **1-year mortality ranging from 25% to 75%** (*N Engl J Med*, PMID: 11794191; *N JACC Heart Fail*, PMID: 27395350). More importantly, as European Society of Cardiology indicated that while the disease trajectory of each patient with HF is unique, there is a generalizable pattern of gradual decline, punctuated by episodes of acute deterioration leading either to sudden death or death due to progressive HF. Thus, anticipatory planning should start to **prevent sudden death when a patient is diagnosed with AHF** (*ESC Scientific Document Group. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure, Eur Heart J*, PMID: 34447992, Subsection 10.1 Epidemiology, diagnosis, and prognosis (page 3637); Subsection 10.2.4 Symptom control and end-of-life care (page 3644)). Furthermore, renal dysfunction (RD) is a major independent determinant of increased mortality and morbidity in HF (*2021 ESC Guidelines*, Subsection 13.6 Kidney dysfunction (page 3669))

(ii) The Mission of Our Prognostic Tool:

To address these challenges, our prediction system was designed to estimate the survival distribution for a given individual with AHF&RD, with ACM serving as primary outcome to train our system. This web-based system (<https://www.hf-ai-survival.com>) offers a

comprehensive range of functionalities for individual patient analysis, including (a) prognostic stratification, (b) estimation of long-term survival probabilities, (c) predictions of all-cause mortality at specific time points, (d) local interpretation, and (e) assessment of predictor contributions.

Adjusting risk predictor values (as identified by the AIHFLevel system) towards their normal ranges through targeted interventions or management strategies can reduce the risk of ACM for this individual, even if the overall prediction categorizes them as high-risk.

(iii) Lack of Other Long-Term Endpoints in External Databases:

After conducting a comprehensive search of the BIDMC database using pgAdmin PostgreSQL Structured Query Language, it was observed that all-cause mortality (ACM) possessed complete event and time variables. In contrast, a majority of other cardiovascular outcome variables were deficient in time data. Consequently, selecting alternative endpoints for the training of our model system would impede external validation, thereby substantially affecting the model's persuasiveness and credibility in a clinical context.

Overall, our goal is to develop an accurate prognostic tool for patients with AHF combined with RD. Given the issues in model development and validation and current clinical challenges, we ultimately decided to train the AIHFLevel prediction system using long-term ACM as the primary endpoint.

(2): We added a Dual-Outcome Evaluation (Sensitivity analysis).

Our aim is to develop a clinically practical tool that is not only precise in performance but also possesses universality and generalizability.

(i) Therefore, in addition to confirming the system's robustness in predicting ACM, we also conducted an evaluation to assess its performance with **major adverse cardiovascular event (MACE)** as an alternative outcome (secondary endpoint). This approach allowed us to

comprehensively gauge the system's predictive capabilities for different clinical endpoints. **(revised manuscript: line 969-981, page 35).**

(ii) Subgroup analysis: We conducted analyses among five pre-specified subgroups [Age (>65 or ≤ 65 years), Cardiac function grade (NYHA I/II or NYHA III/IV), HF subtype (HFpEF, HFmrEF, HFrEF), CKD stage (stage I/II, III, IV/V), PCI history (Yes or No)] to evaluate the prognostic value of AIHFLevel on long-term ACM and MACE among AHF patients with RD. **(revised manuscript: line 969-981, page 35).**

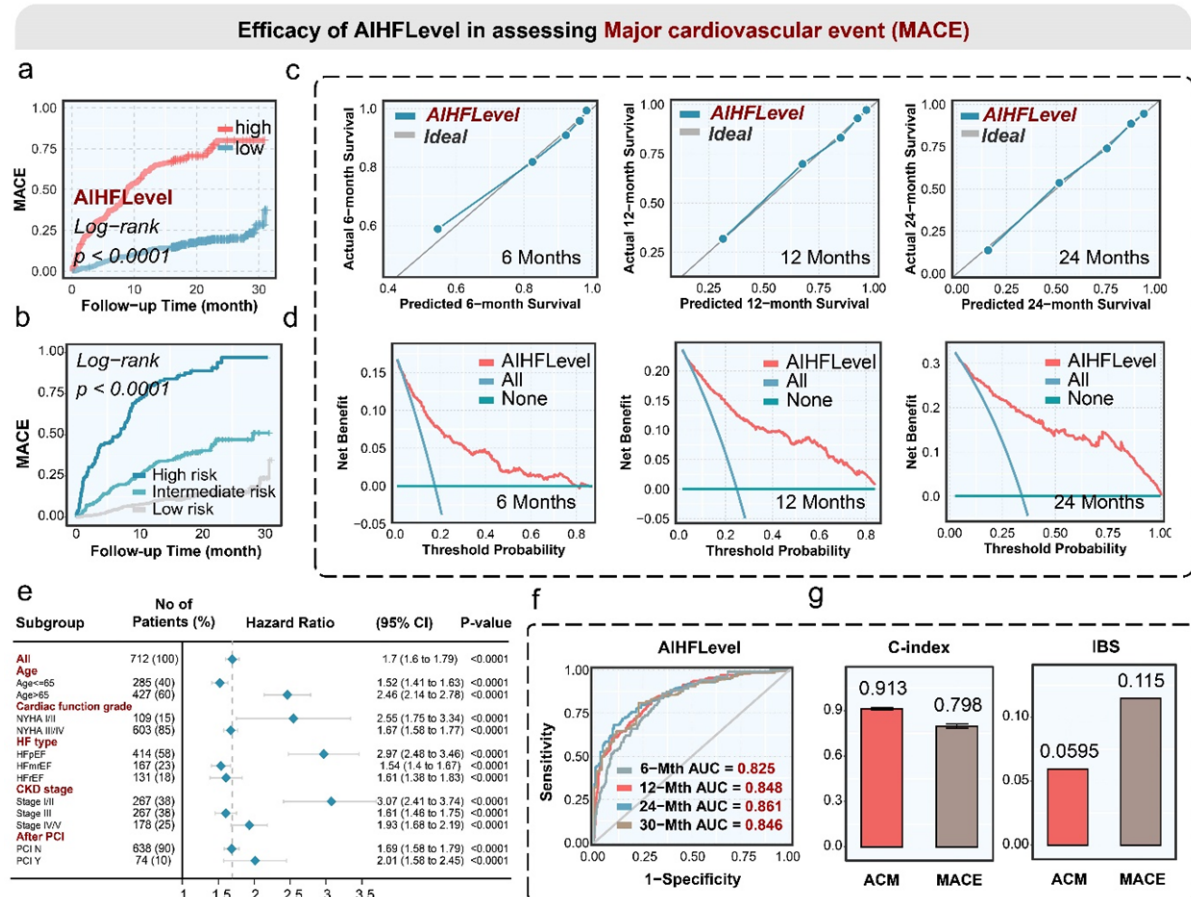
(iii) Define MACE: MACE included one or several of the following conditions: readmitted for symptomatic HF, nonfatal acute coronary syndrome, nonfatal ischemic stroke, new-onset cardiac arrhythmia, the use of mechanical circulatory support and implementation of heart transplantation, and cardiac death. We reviewed each chart separately to determine the presence of MACE, and the American College of Cardiology/American Heart Association definitions provided for clinical trials were used to identify MACE (*J Am Coll Cardiol*, PMID: 25553722) **(revised manuscript: line 773-780, page 28).**

Sensitivity analysis: (revised manuscript 'Result section': line 316-335, page 12-13).

To enhance the robustness assessment of AIHFLevel, we expanded the evaluation to include major adverse cardiovascular events (MACE) as an alternative outcome based on the Meta cohort. Initially, stratifying patients into high and low AIHFLevel groups based on median values, the Kaplan–Meier curves revealed a significantly higher rate of MACE in the high AIHFLevel group ($P<0.0001$) (Fig. 5a). Significant risk differences for MACE were also observed across the three prognostic states ($P<0.0001$) **(Revised Fig. 5b)** Calibration curves indicated similar predicted and actual probabilities for MACE **(Revised Fig. 5c)**. DCA analysis further demonstrated the predictive potential for MACE **(Revised Fig. 5d)**. Stratification analysis demonstrated the AIHFLevel's consistent prognostic value for MACE across different subgroups **(Revised Fig. 5e)**. ROC analysis confirmed AIHFLevel's discriminative ability for MACE, and AUCs at 6-, 12-, 24-, and 30-month were respectively 0.825, 0.848, 0.861, 0.846 **(Revised Fig. 5f)**. Notably, AIHFLevel demonstrated enhanced accuracy in predicting ACM over MACE (MACE: C-index=0.798, IBS=0.115; ACM: C-index=0.913, IBS=0.0595) **(Revised Fig. 5g)**.

Designed to estimate individual survival distributions, AIHFLevel was initially trained with ACM as its primary endpoint. However, when evaluated against MACE as an alternative endpoint, it still demonstrated reliable predictive performance. The ability to accurately assess

outcomes across different clinical endpoints indicated AIHFLevel's comprehensive predictive capabilities and robustness in clinical prognostication.



Revised Fig. 5: Sensitivity analysis for AIHFLevel.

a Cumulative Kaplan–Meier estimates of the time to the first adjudicated occurrence of MACE within the Meta cohort, with log-rank test results indicating statistical significance.

b Kaplan-Meier curve showing the incidence difference for MACE across three prognostic states, with log-rank test results indicating statistical significance.

c Calibration curve depicting the predicted versus observed probabilities of MACE at 6, 12, and 24 months, as evaluated by AIHFLevel within Meta cohort.

d DCA illustrating net benefit curves of AIHFLevel for predicting MACE at 6, 12, and 24 months within Meta cohort. The X-axis represented the threshold probability for critical care outcomes, while the Y-axis quantified the net benefit.

e Subgroup analysis estimating AIHFLevel's prognostic value for MACE across different subgroups. The length of the horizontal line represented the 95% confidence interval for each subgroup.

subgroup, with a vertical dotted line indicating the hazard ratio of all patients. The vertical solid line denoted HR=1. HR>1 indicated AIHFLevel as a risk prognostic factor.

f Time-dependent ROC analysis for predicting MACE within Meta cohort. AUCs at 6-, 12-, 24-, and 30-months demonstrating strong predictive accuracy: 0.825, 0.848, 0.861, 0.846.

g Comparative AIHFLevel's efficacy in assessing ACM and MACE within the Meta cohort using C-index and IBS.

The revised content mentioned above has been incorporated into the 'Study outcome' subsection of the Methods section (**revised manuscript: line 766-780, page 28**), ' Subgroup and sensitivity analyses' subsection of the Methods section (**revised manuscript: line 969-981, page 35**), and the ' Sensitivity analysis' subsection of the Results section (**revised manuscript: line 316-335, page 12-13**).

Once again, we sincerely appreciate your insightful suggestions, which we're confident will greatly improve the robustness, reliability of our research. We genuinely hope for your support and endorsement!

Q3: It is not entirely clear which particular outcome has been used to implement a consensus assessment strategy. Was it all-cause mortality across the whole time span? Which was then assessed in various subgroups; is that correct?

Response:

We are grateful for your thoughtful feedback and attention to this matter. We apologize for any confusion caused by the previous manuscript. We have enhanced the revised manuscript with more detailed textual descriptions and illustrative figures. We sincerely hope these revisions meet your expectations.

To clarify, the primary endpoint for our consensus assessment strategy was ACM across the entire time span. Secondly, such strategy was designed to pick out the optimal modeling scheme for AHF&RD patients, not involved the subsequent internal and external validation processes for the prediction system. Briefly, through our framework, we conducted feature preprocessing, generated modeling schemes, selected the optimal scheme, and established the prediction model. Subsequently, this prediction system was initially tested in various subgroups of the CRCCD cohort (internal testing datasets) and then validated in the BIDMC cohort (external validation datasets) to ensure the robustness and generalizability of our system.

Given that such consensus assessment strategy as one crucial component of our AI hybrid modeling framework, we would like to provide a more organized and clear presentation of the entire framework.

(1): Why we committed to develop an AI hybrid modeling framework? (Revised manuscript line 782-794 page 28-29).

There are many modeling algorithms to analyze data and assess outcomes currently. Predictions provided by different algorithms may vary depending on the characteristics of the dataset. It remained challenging to identify the appropriate learner tailored to the application's

requirements. Despite many options available, researchers' preferences and knowledge limitations can lead to suboptimal use of modeling algorithms, resulting in reduced predictive power (*J Intern Med*, PMID: 35426190). Algorithm selection should be based on various objectives, conditions, aligned with known rules (*Cardiovasc Diabetol*. PMID: 36658644).

However, some researchers have suggested that the use of hybrid learning models, e.g. ensemble-based approaches that is, a combination of two models, and enhancing existing learning models can be helpful. Mienye and colleagues provided evidence that ensemble-based learning may perform better than the single model in the prediction of HF events (*Informatics in Medicine Unlocked*, <https://doi.org/10.1016/j.imu.2020.100402>). Ensemble modeling operates by employing a collection of diverse and independent models to make predictions, thereby reducing the generalization error and improving the reliability of the predictive outcomes (*AMIA Annu Symp Proc*, PMID: 30815108).

(2): How dose it work? (Revised manuscript line 796-886, page 28-32).

The graphic illustration of the workflow was presented in **(Revised Fig. 2a, Revised Supplementary Fig. 9)**

(I) Dataset: The CRCCD in-house cohort was used to evaluate the viability of various algorithms and identify the optimal modeling schemes, while the BIDMC external cohort served to validate the system's generalizability.

ACM serves as an outcome to train our system.

(II) Variable preprocessing: We incorporated 93 readily accessible variables, as detailed earlier **(Revised manuscript line 669-710, page 24-25)**. In the CRCCD cohort, univariate Cox proportional hazards (PH) regression and Log-rank test were conducted. 46 variables were identified as latent survival features, demonstrating statistical significance ($P < 0.05$) in both tests.

(III) 12 modeling algorithms: L1-regularized PH regression (L1.regularization), L2-regularized PH regression (L2.regularization), Elastic-Net-regularized PH regression (ENet.regularization), Survival random forest SRC learner (Surv.RFsrc), Survival support vector machine (Surv.SVM), survival tree (Surv.rpart), Cox model with likelihood-based boosting (Surv.Coxboost), Boosted generalized linear survival learner (Surv.glmboost), Extreme gradient boosting survival learner (Surv.Xgboost), Survival gradient boosting machine learner (Surv.gbm), Survival fully parametric learner (Surv.Fully-parametric), Accelerated oblique random survival forest learner (Surv.aorsf). Details of their hyperparameters are presented in **(Revised Supplementary Table 20)**.

(IV) 132 modeling schemes:

(a) Single Algorithm Model Fitting: 12 algorithms were performed on 46 latent survival features to fit models, generating 15 schemes: L1.regularization (lambda.min), L1.regularization (lambda.1se), L2.regularization (lambda.min), L2.regularization (lambda.1se), ENet.regularization (lambda.min), ENet.regularization (lambda.1se), Surv.RFsrc, Surv.SVM, Surv.rpart, Surv.Coxboost, Surv.glmboost, Surv.Xgboost, Surv.gbm, Surv.Fully-parametric, Surv.aorsf. For regularization algorithms (L1, L2, and ENet), we utilized 'lambda.min' and 'lambda.1se' as λ regularization parameters to expand the range of possible schemes.

(b) Model Fitting using a Combinatorial Family of Algorithms: Feature selection on 46 latent survival features was initially executed with each algorithm, followed by fitting the selected features with the remaining algorithms.

Embedded Method: For L1.regularization and ENet.regularization, we directly leveraged their built-in selection mechanisms based on L1 and elastic-net penalties. 5-fold cross-validation served as the resampling strategy for evaluating feature subset performance. The other 9 algorithms, excluding regularization methods, were then applied to fit models on the selected features, generating 18 schemes.

Filter & Wrapper Hybrid Method: This method combined the rapid training capability of the external technique Filter (feature ranking) and the intensive computation of the Wrapper (using the model as a black box to evaluate feature subset according to their predictive power) (*A survey on feature selection methods, Computers & Electrical Engineering, <https://doi.org/10.1016/j.compeleceng.2013.11.024>*). Apart from the regularization methods, each of the remaining 9 algorithms executed the following steps: First, the filter method was adopted to rank the features in descending order according to the Correlation-Adjusted Survival Score (CARS). Next, sequential forward floating selection (SFFS) as a wrapper method was used to select features. Features were added in feature sets in sequence, and feature retention was based on the performance of each algorithm at every step. The 5-fold cross-validation assisted the performance assessment. If the performance with a new feature set did not increase, the new feature was removed from the feature set. **Revised Supplementary Fig. 10** illustrated the main procedure of the Filter & Wrapper Hybrid Method. Finally, the remaining 11 algorithms were used to fit models on the selected features, generating 99 schemes.

(V) Consensus assessment strategy:

In the CRCCD cohort, we implemented 132 modeling schemes, each generating a separate model. The evaluation of these models' effectiveness and relevance was conducted via a rigorous comprehensive assessment strategy. This included:

- 1. 10 repeated 10-fold cross-validation,**
- 2. Monte-Carlo cross-validation (MCCV, 100 iterations with a 0.7 sampling ratio),**
- 3. bootstrap analysis (1000 iterations with a 0.7 sampling ratio).**

Specifically,

1. 10 repeated 10-fold cross-validation:

In this K-Fold cross-validation technique, the entire dataset (CRCCD cohort) is divided into K equal-sized parts. Each partition is referred to as a "Fold." Thus, we have K Folds, and we call it K-fold. One fold is used as the validation set, while the remaining K-1 folds are used as the training set. We repeated this process for 10 times. Take the average of all the evaluation results as the performance evaluation of the model.

[figure redacted]

2. Monte-Carlo cross-validation (MCCV, 100 iterations with a 0.7 sampling ratio):

This method addresses the shortcomings of traditional cross-validation methods. Its basic idea is to randomly split the dataset into two subsets: a training set and a validation set. This process is repeated multiple times, with different training and validation sets each time, and the average of the results is used as the performance evaluation of the model:

Randomly split the dataset into a training set and a validation set;

Train the model using the training set;

Evaluate the model's performance using the validation set;

Repeat steps 1-3 multiple times (e.g., 100 times), with different training and validation sets each time.

Take the average of all the evaluation results as the performance evaluation of the model. MCCV can fully utilize the information in the dataset, reduce the variance of the model, and improve the accuracy of the model's evaluation. However, it requires a large amount of computation and may take a longer time to complete the evaluation.

[figure redacted]

3. bootstrap analysis (1000 iterations with a 0.7 sampling ratio):

Bootstrap, also known as bootstrapping, is a resampling technique that involves sampling with replacement from the original dataset (CRCCD cohort) to create new datasets of the same size. These new samples are used as training sets, while the original dataset is used as a validation set to evaluate model performance. This process is repeated 100 times. Take the average of all the evaluation results as the performance evaluation of the model.

[figure redacted]

The concordance index (C-index), calculated across three strategies, served as the pivotal performance metric. The modeling scheme with the highest average C-index was recognized as the most effective for AHF&RD patients.

This consensus assessment strategy offered **advantages**:

(a) **Comprehensive Coverage:** Each assessment method has both advantages and disadvantages. Integrating multifaceted validation techniques ensured a more thorough and resilient evaluation. This multifaceted approach countered the risks of overfitting, bias inherent in single-method validation and selection bias by researchers.

(b) **Diverse Scenario Analysis:** Using MCCV and bootstrap analysis allowed for an extensive examination of the models' performance across diverse datasets and hypothetical clinical scenarios, thus ensuring that the model is not overly tailored to a specific dataset or condition. This enhances the model's generalizability, making it more applicable and reliable in different clinical settings and populations.

(c) **Precision in Model Selection:** Applying the average C-index across diverse validation strategies refined the selection of the most effective model for AHF&RD, ensuring a robust choice backed by consensus performance metrics.

(d) **Reduction of Variability:** The repeated application of validation techniques, such as 10-fold cross-validation and bootstrap analysis, helps to reduce the variability in the model's performance estimates. This leads to more stable and reliable model evaluations, minimizing the impact of random fluctuations in the data.

(VI) Validation and Extrapolation: In the CRCCD in-house cohort, we divided the participants into a Discovery cohort (70%, n = 498) for model fitting using the optimal scheme,

which led to the modeling of our prediction system AIHFLevel. A Replication cohort (30%, n = 214) was then employed for the initial validation of AIHFLevel. Further validation and generalization processes were conducted on the complete CRCCD Meta cohort and the external BIDMC cohort. External validation across diverse data modalities, such as patients from BIDMC with varying hospital systems, technological platforms, regional, and ethnic backgrounds, is crucial for predictive analytics. This approach can effectively address the issue of model overfitting on internal data, ensuring broader applicability and generalizability of the predictive system.

In summary, as highlighted by TRIPOD statement, studies developing new prediction models should include diverse form of internal validation to quantify any optimism in the predictive performance of the developed model (*Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD): explanation and elaboration, PMID: 25560730*, Subsection ‘Prediction model studies’ (page 2)).

Our consensus assessment strategy allows for the continuous refinement of the modeling schemes based on its performance across different validation techniques. This adaptive approach enables the identification and correction of any weaknesses in the model, leading to the most accurate and robust predictive model for AHF&RD patients.

We hope these presentations and revisions meet your expectations and contribute to the study's rigor. Your kind and professional guidance is highly appreciated!

Q4: With regards to the data, it would be beneficial if the authors could provide more details on the imputation. Is the missing at random mechanism justified? How much data was missing, and for which parameters? How did the missingness vary across the training and testing sets? With the biomarkers, for example, frequently the data might be "missing" depending on the patient's state - those who are "healthier" would have fewer examinations. Please add a bit more detail to that.

Response:

We value your comments on missing data. Our initial manuscript briefly outlined our approach, but lacked detail. We apologize for any confusion. In the revised manuscript, we've provided a detailed account of the missing data extent and our handling methods

(1) Variables collection:

We retrieved the patient data within the initial 24 hours following admission. 93 easily accessible variables were listed as candidates for constructing the survival assessment system subsequently. Further details can be found in the 'Data collection' subsection of the Methods section (**revised manuscript: lines 669-710, pages 24-25**).

(2) Missingness information:

The TRIPOD guideline suggested applying multiple imputations when missing data were present, as complete case analyses can lead to reduced sample size, biased estimates, and loss of Information. Variables with over 20% missing values should be excluded in the following analyses to minimize the bias resulting from missing data. (*Transparent Reporting of a multivariable prediction model for Individual Prognosis or Diagnosis (TRIPOD): explanation and elaboration. Ann Intern Med. PMID: 25560730; Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. BMJ. PMID:*

19564179). The BIDMC cohort presented no instances of missing data, while the extent of missingness in the CRCCD cohort is detailed in (revised Supplementary Table 19).

Supplementary Table 19. Missingness report for Henan Province Clinical Research Center for Cardiovascular Diseases Cohort.

	Missing number	Total number	Missing percentage	Variable type
NYHA	93	712	13.06%	Categorical
CKD stage	27	712	3.79%	Categorical
Fg	1	712	0.14%	continuous
BNP	1	712	0.14%	continuous
CRP	4	712	0.56%	continuous
GLU	2	712	0.28%	continuous
RVID	9	712	1.26%	continuous
IVS	10	712	1.40%	continuous
LVID	8	712	1.12%	continuous
LVPWT	8	712	1.12%	continuous
AOAD	9	712	1.26%	continuous
LAD	8	712	1.12%	continuous
PVAD	9	712	1.26%	continuous
RALRD	43	712	6.04%	continuous
RASID	54	712	7.58%	continuous
MVE	72	712	10.11%	continuous
MVA	82	712	11.52%	continuous
EF	13	712	1.83%	continuous
SV	44	712	6.18%	continuous
FS	44	712	6.18%	continuous

Abbreviations:

NYHA, New York Heart Association Classification System; CKD, chronic kidney disease; CRP, C-reactive protein; GLU, glucose; RVID, right ventricle internal diameter; IVS, interventricular septal thickness; LVID, left ventricle internal diameter; LVPWT, left ventricular posterior wall thickness; AOAD, aortic valve annulus diameter; LAD, left atrium diameter; PAVD, pulmonary artery valve diameter; RALRD, right atrium lateral diameter; RASID, right atrium supero-

inferior diameter; MVE, early diastolic mitral wave velocity; MVA, late diastolic mitral wave velocity ratio; EF, ejection fraction; SV, stroke volume; FS, left ventricular fraction shortening

(3) Patterns of Missingness:

Rubin defined the mechanisms of missing data as follows: Missing Completely at Random (MCAR), Missing at Random (MAR), and Missing Not at Random (MNAR) (*Donald B. Rubin (1986) Statistical Matching Using File Concatenation With Adjusted Weights and Multiple Imputations, Journal of Business & Economic Statistics, 4:1, 87-94, DOI: 10.1080/07350015.1986.10509497*).

(I) Missing Completely at Random (MCAR): The reason for the missing data is independent of both observed and unobserved data. In other words, the missingness is unrelated to any variables, whether observed or not. The missing data occur randomly. Every record in any variable has the same probability of being missing. MCAR is a very strict assumption and is rare in practice. (Example: Data loss due to entry errors or instrument failure).

(II) Missing at Random (MAR): After controlling for other variables, the missing data are unrelated to the variable itself but are related to other study variables. In other words, the missing data can be predicted using other observed values. If we can fully observe the variables related to the missingness, we can achieve a dataset that is missing at random.

(III) Missing Not at Random (MNAR): Excludes MCAR and MAR. The missing data are related to the variable itself and depend on its hypothetical values. It can also be understood that the missing data are likely to be related to the characteristics of the unobserved data, leading to bias. Example: Deliberately concealing income information. Even if other observed values are included in the statistics, it is difficult to infer the missing income information (hidden by the individual's unwillingness to disclose).

(4) Missingness handling:

The approach to handling missing data involves identifying the missing values through certain methods, analyzing the proportion of missing values in the overall dataset, and examining whether there are significant patterns of random distribution. Subsequently, we decide which method to use for missing data imputation.

The purpose of handling missing values is not to accurately estimate the true values of the missing data, but to predict the distribution that the missing values follow. This allows for prediction (inference of prediction equations and model evaluation), making these model parameters effective for external data.

The methods for handling missing data mainly fall into two categories: "deletion" and "imputation," depending on the assumptions about the missing data mechanism. For example, deleting subjects with missing values and mean-value imputation are generally used for MCAR, while common imputation methods in the literature, such as regression imputation and multiple imputation, are based on the MAR assumption. For MNAR, there are currently no effective methods (*Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. BMJ, PMID: 19564179; Missing Data in Clinical Research: A Tutorial on Multiple Imputation. Can J Cardiol, PMID: 33276049*)

The TRIPOD guideline suggested applying **multiple imputations** when missing data were present, as complete case analyses can lead to reduced sample size, biased estimates, and loss of Information. Variables with over 20% missing values should be excluded in the following analyses to minimize the bias resulting from missing data.

In general, multiple imputation involves the following two steps:

(a) Creating Datasets:

Multiple copies of the dataset are created, with missing values replaced by imputed values. The imputed values are calculated based on the observed data to predict the overall distribution of the data and are then sampled. Multiple imputation is based on Bayesian methods. Due to the uncertainty of the prediction, multiple imputations are usually performed to generate several imputed values (introducing appropriate variability) for prediction. Typically, imputation is done five times.

(b) Statistical Analysis:

The same statistical method is applied to each imputed dataset (the same as used for a single dataset). Theoretically, the estimated values and parameters from the statistical analysis of each dataset will differ. Therefore, it is meaningful to combine them by averaging the distribution of missing values to provide an overall association. This process should follow Rubin's rules.

(5) Multiple imputations:

Initially, our data was presumed to be Missing At Random (MAR), aligning with the assumptions of prior studies (*Eur J Heart Fail. PMID: 30919549; JAMA Netw Open. PMID: 34767024*). We conducted Little's test to confirm that the data were not missing completely at random (MCAR) (*A Test of Missing Completely at Random for Multivariate Data with Missing Values, Journal of the American Statistical Association, doi:10.1080/01621459.1988.10478722*). The result was depicted in the (revised **Supplementary Table 19**). Therefore, multiple imputation was performed through the method of chained equations procedure, ensuring a statistically robust approach to addressing missing data in our study (*Multiple imputation by chained equations: what is it and how does it work? Int J Methods Psychiatr Res. PMID: 21499542*)

Supplementary Table 19. Use Little's test to assess for missing completely at random (MCAR), suggesting our data is unlikely to be MCAR

Little's test	Chi.square	(Degrees of freedom used for chi-square)	P-value
value	1483.688	1104	1.26e-13

Little, R. J. A. (1988). A test of missing completely at random for multivariate data with missing values. *Journal of the American Statistical Association*, 83(404), 1198–1202.

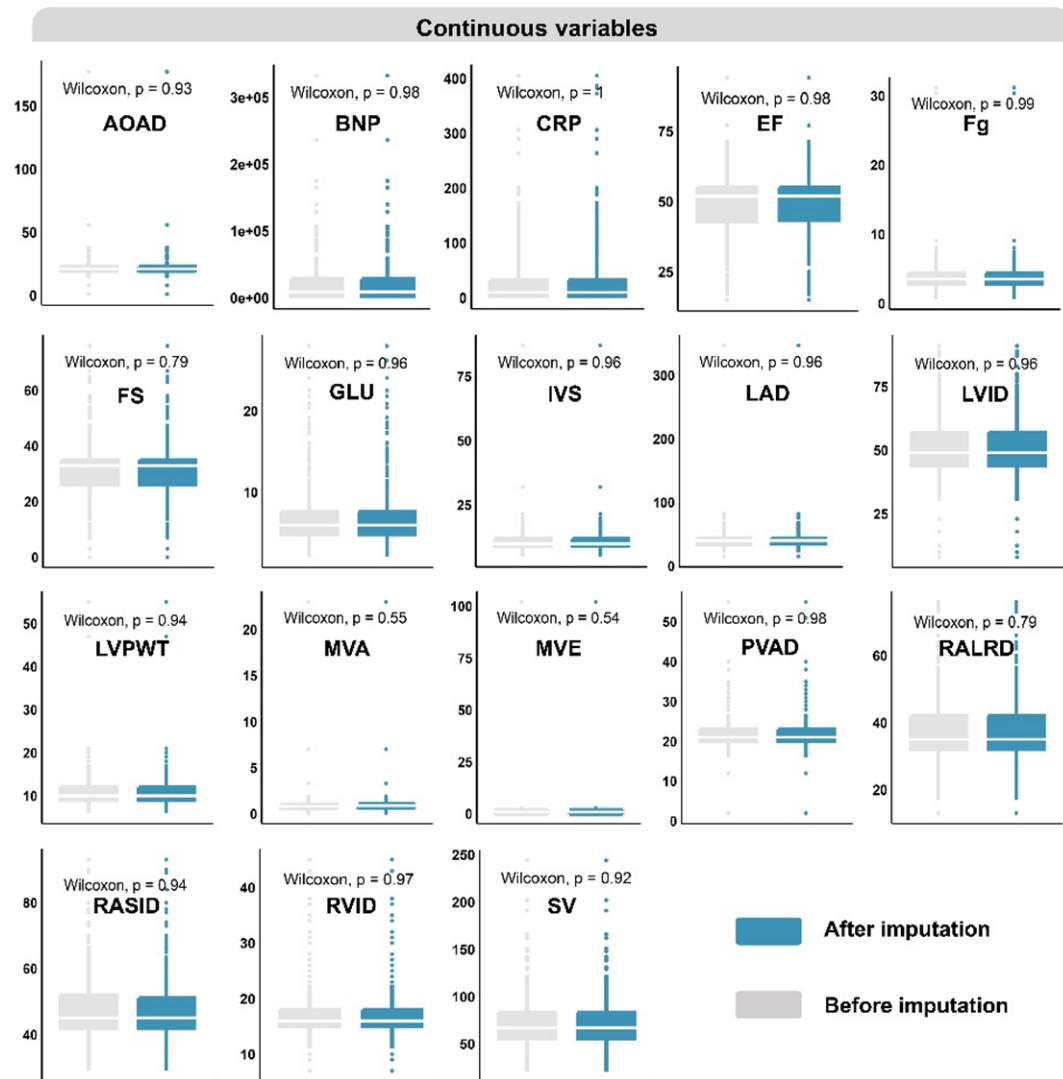
Multiple imputation with chained equations (MICE) specified a multivariate model through conditional distributions, where each variable with missing data is imputed conditionally on all other variables. In our dataset with p (p=93) variables and k (k=20) variables missing data (with p-k complete cases), the procedure was as follows:

- (1) For each of the k variables with missing data, an imputation model was specified.
 - (a) numeric data: bayesian linear regression imputation.

- (b) factor data with 2 levels: logistic regression imputation.
 - (c) factor data with >2 unordered levels: polytomous regression imputation.
 - (d) factor data with >2 ordered levels: proportional odds model.
- (2) Imputation values are drawn from the overall distribution of observed values for each variable with missing data. These initial imputed values are refined in subsequent steps.
- (3) For the first variable with missing data:
- (a) Regression is performed based on other variables (including the complete data of the first variable; and other observed or imputed variables).
 - (b) Regression parameters: extract estimated regression coefficients and the variance-covariance matrix of the regression model from the previous regression (and for linear regression models fitted for continuous variables, the estimated variance of the residuals).
 - (c) Parameters are randomly perturbed to reflect the uncertainty in data generation. The conditional distribution for each sample with missing data in the first variable is determined using the perturbed regression coefficients.
 - (d) Imputation values are then selected for each missing datum based on the conditional distribution.
- (4) Steps 3 and 4 are repeated in a cycle, imputing each variable with missing data.
- (5) The process of steps 3 and 4 is repeated 20 iterations. The final imputed values form the first completed imputed dataset.
- (6) Steps 3 to 5 are repeated 5 times to generate 5 imputed datasets.
- (7) Construction of the multiple imputed datasets is completed and now M datasets are obtained (assuming M=5). These datasets are then treated as complete for statistical analysis. The results were pooled into one dataset by applying Rubin's rules.

(3) Reliability of Missing Data Handling

In addressing missing data for continuous and categorical variables, our analysis revealed a uniform pattern in the data distribution before and after interpolation, as detailed in **(revised Supplementary Fig. 7)** and **(revised Supplementary Fig. 8)**. This consistency was validated using the Wilcoxon rank-sum test for continuous variables and the chi-squared test for categorical variables, ensuring the integrity and reliability of our data analysis.



Supplementary Figure 7. In the process of managing missing continuous variable data within the in-house cohort (Henan Provincial Cardiovascular Disease Clinical Research Center), we observed a consistent data distribution pattern both pre- and post-interpolation. The comparison was assessed using the Wilcoxon rank-sum test.

Categorical variables					
	After imputation	Before imputation	Pearson's Chi-squared test X-squared	Pearson's Chi-squared test df	Pearson's Chi-squared test P-value
Cohort size	N = 712	N = 712			
NYHA (%)	n = 712 (100.00(%))	n = 619 (86.94(%))	2.1286	3.0	0.5461
IV	26 (3.65%)	16 (2.58%)			
III	86 (12.08%)	67 (10.82%)			
II	225 (31.6%)	193 (31.18%)			
I	375 (52.67%)	343 (55.41%)			
CKD stage (%)	n = 712 (100.00(%))	n = 685 (96.21(%))	2.3321	5.0	0.8015
V	168 (23.6%)	168 (24.53%)			
IV	136 (19.1%)	135 (19.71%)			
IIIb	128 (17.98%)	127 (18.54%)			
IIIa	22 (3.09%)	13 (1.9%)			
II	249 (34.97%)	233 (34.01%)			
I	9 (1.26%)	9 (1.31%)			

Supplementary Figure 8. In the process of managing missing categorical variable data within the Henan Provincial Cardiovascular Disease Clinical Research Center cohort, we observed a consistent data distribution pattern both pre- and post-interpolation. The comparison was assessed using the chi-squared contingency table tests.

Based on your valuable recommendations, we have elaborated further on this aspect in the Missing data handling' subsection of the Methods section (**revised manuscript: lines 712-764, pages 26-28**).

We appreciate your constructive feedback, which has greatly enhanced the clarity and depth of our study. We trust that these revisions align with your expectations and comprehensively address your concerns.

Q5: The average age in the external cohort is much higher than in the training cohort. Also, the all-cause mortality is much higher. Could the authors expand on that, please, and discuss why they believe this should not impact the generalisability of the results?

Response:

We appreciate your thoughtful feedback and attention to this matter. In response, we have enriched the revised manuscript with more detailed information. We sincerely hope that these revisions meet your expectations and address your concerns effectively.

Answer to the question 'The average age in the external cohort is much higher than in the training cohort. Also, the all-cause mortality is much higher. Could the authors expand on that, please'.

In-depth baseline data for CRCCD and BIDMC cohorts, including comorbidity status, blood routine, coagulation, renal and liver functions, cardiac assessments, echocardiography results, medication history, were detailed in (**Supplementary Tables 1 and 2**). Based on your kind suggestions, we further compared the important baseline characteristics of the two cohorts, as depicted in (**Revised Supplementary Table 14**).

Supplementary Table 14. Comparisons of baseline characteristic data between the in-house cohort from Henan Province Clinical Research Center for Cardiovascular Diseases (CRCCD) and external cohort from Beth Israel Deaconess Medical Center (BIDMC).

Characteristic	Cohort			p-value ²
	Overall, N = 1,736 ¹	CRCCD, N = 712 ¹	BIDMC, N = 1,024 ¹	
ACM time	19 (16)	19 (8)	19 (20)	<0.001
ACM				<0.001
Alive	1,020 / 1,736 (59%)	561 / 712 (79%)	459 / 1,024 (45%)	
Dead	716 / 1,736 (41%)	151 / 712 (21%)	565 / 1,024 (55%)	
Age	66 (16)	59 (16)	71 (14)	<0.001
Arrhythmia				<0.001

Co-morbidity				
N	887 / 1,736 (51%)	484 / 712 (68%)	403 / 1,024 (39%)	
Y	849 / 1,736 (49%)	228 / 712 (32%)	621 / 1,024 (61%)	
CAD				0.8
Co-morbidity				
N	886 / 1,736 (51%)	366 / 712 (51%)	520 / 1,024 (51%)	
Y	850 / 1,736 (49%)	346 / 712 (49%)	504 / 1,024 (49%)	
GFR, %	56 (34)	52 (26)	59 (39)	0.007
LY_Per, %	16 (10)	17 (9)	14 (11)	<0.001
TBIL, µmol/L	6 (14)	14 (20)	1 (2)	<0.001
TN, ng/mL	0.60 (2.44)	0.50 (2.52)	0.68 (2.39)	<0.001
CKD_stage				0.005
I II	730 / 1,736 (42%)	267 / 712 (38%)	463 / 1,024 (45%)	
IIIa IIIb	595 / 1,736 (34%)	267 / 712 (38%)	328 / 1,024 (32%)	
IV V	411 / 1,736 (24%)	178 / 712 (25%)	233 / 1,024 (23%)	
Cr, µmol/L	268 (306)	414 (385)	167 (175)	<0.001
SV, mL	67 (37)	71 (25)	64 (43)	<0.001
MCHC, g/L	331 (25)	326 (30)	334 (20)	<0.001
EF, %	54 (18)	48 (10)	58 (21)	<0.001
Gender				0.003
F	750 / 1,736 (43%)	277 / 712 (39%)	473 / 1,024 (46%)	
M	986 / 1,736 (57%)	435 / 712 (61%)	551 / 1,024 (54%)	
WBC, 10⁹/L	10.2 (17.9)	7.7 (3.7)	12.0 (22.9)	<0.001
RBC, 10⁹/L	3.78 (0.84)	3.60 (0.94)	3.90 (0.74)	<0.001
Hb, g/L	53 (61)	111 (56)	12 (2)	<0.001
Plt, 10⁹/L	227 (106)	195 (87)	249 (113)	<0.001
MCV, fL	90 (9)	90 (12)	90 (7)	<0.001
MCH, pg	30.29 (8.40)	30.73 (12.66)	29.99 (2.87)	0.4
RDW, %	16.16 (13.85)	17.38 (21.43)	15.31 (2.11)	0.2

PT, s	29 (24)	18 (16)	37 (25)	<0.001
Fg, g/L	3.79 (2.01)	3.70 (1.83)	3.86 (2.12)	0.3
BUN, mmol/L	9 (11)	18 (11)	2 (2)	<0.001
UA, mmol/L	532 (262)	411 (169)	616 (282)	<0.001
TP, g/L	63 (16)	66 (20)	61 (11)	<0.001
TCHO, mmol/L	5.83 (2.65)	3.97 (2.03)	7.13 (2.22)	<0.001
HDL, mmol/L	1.64 (0.93)	1.03 (0.57)	2.06 (0.89)	<0.001
LDL, mmol/L	3.23 (1.69)	2.29 (1.15)	3.89 (1.69)	<0.001
NT-proBNP, ng/mL	13,969 (20,641)	17,515 (25,992)	11,504 (15,434)	<0.001
CK, U/L	287 (2,236)	126 (189)	398 (2,902)	<0.001
CRP, mg/L	52 (67)	27 (49)	69 (72)	<0.001
GLU, mmol/L	7.9 (4.7)	7.0 (3.7)	8.6 (5.1)	<0.001
HF subtype				<0.001
HFmrEF	285 / 1,736 (16%)	167 / 712 (23%)	118 / 1,024 (12%)	
HFpEF	1,099 / 1,736 (63%)	414 / 712 (58%)	685 / 1,024 (67%)	
HFrEF	352 / 1,736 (20%)	131 / 712 (18%)	221 / 1,024 (22%)	
TRIG, mmol/L	85 (102)	2 (1)	143 (97)	<0.001
ALB, g/L	17 (18)	37 (13)	3 (1)	<0.001
¹ Mean (SD); n / N (%)				
² Wilcoxon rank sum test; Pearson's Chi-squared test				

Indeed, apart from a few indicators with insignificant differences, the majority of differences between the two cohorts are notably significant. This observation underscores the heterogeneity of patients from different centers and affirms the independence of the two cohorts. As noted in one study, there is substantial lack of data from developing countries, where HF exhibits different features compared with that observed in the Western world (*Global burden of heart failure: a comprehensive and updated review of epidemiology. Cardiovasc Res. 2023 Jan 18;118(17):3272-3287. PMID: 35150240*).

For the difference of the age and ACM, we conducted further investigation based on your insightful suggestions. As noted in *the 2021 Guidelines*, in developed countries, the age-adjusted incidence of HF may be falling, presumably reflecting better management of cardiovascular disease, but due to ageing, the overall incidence is increasing. Due to ageing and the increasing prevalence of comorbidities, the absolute number of hospital admissions for HF is expected to increase considerably in the future, perhaps by as much as 50% in the next 25 years (*2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure, Eur Heart J, PMID: 34447992, Section 3.3 Epidemiology and natural history of heart failure, Subsection 3.3.1 Incidence and prevalence (page 3614)*). Furthermore, *the 2021 Guidelines* also noted that the prevalence of AHF is increasing due to the growing number of patients with HF, ageing of the population in Western countries (*Section 10 Advanced heart failure, Subsection 10.1 Epidemiology, diagnosis, and prognosis (page 3637)*). A population-based study of 4 million individuals have reported that over the study period, patient age and multi-morbidity at first presentation of HF increased in the Western world (mean age 76.5 years [SD=12.0] to 77.0 years [12.9], adjusted difference 0.79 years, 95% CI 0.37-1.20 (*Temporal trends and patterns in heart failure incidence: a population-based study of 4 million individuals. Lancet, PMID: 29174292*)). The China Heart Failure (China-HF) registry is an ongoing prospective, multicenter, and largest-scale registry study of hospitalized heart failure patients in China. The study found that the average age of heart failure patients in China is 62 years old, which is at least 6 years younger than in developed countries such as Europe and America (average age >70 years). Researchers suggested that this may be related to poorer control of heart failure risk factors, delayed diagnosis, or poor medication adherence among patients with coronary heart disease, leading to premature onset (*China-HF Investigators. Contemporary Epidemiology, Management, and Outcomes of Patients Hospitalized for Heart Failure in China: Results From the China Heart Failure (China-HF) Registry. J Card Fail. PMID: 29029965*). These evidences partially explains the age differences between the two cohorts (CRCCD: mean age = 59 years [SD = 16]; BIDMC: mean age = 71 years [SD = 14]).

Regarding ACM, the higher incidence in the BIDMC cohort during the study period is partly due to their longer follow-up time (nearly 50 months), resulting in a higher number of ACM events, while the maximum follow-up time in the CRCCD cohort was 30 months. Moreover, higher country- and patient-level socio-economic indicators were associated with higher quality of life and lower risk of all-cause mortality/HF hospitalization (*Global, regional, and national incidence, prevalence, and years lived with disability for 354 diseases and*

injuries for 195 countries and territories, 1990-2017: a systematic analysis for the Global Burden of Disease Study 2017. Lancet. PMID: 30496104). The GWTG-HF registry analyzed 39 982 patients hospitalized for HF between 2005 and 2009 in the USA.¹⁰⁹ Median survival was 2.1 years and 5-year mortality was 75.4%. (*Heart Failure With Preserved, Borderline, and Reduced Ejection Fraction: 5-Year Outcomes. J Am Coll Cardiol. PMID: 29141781*). In the ASIAN-HF registry, crude 1-year all-cause mortality was 12.6% in patients with symptomatic HF, which corroborated our study that 10-month ACM was 15 (*Regional Variation of Mortality in Heart Failure With Reduced and Preserved Ejection Fraction Across Asia: Outcomes in the ASIAN-HF Registry. J Am Heart Assoc. PMID: 31852421*).

Answer to the question ' **discuss why they believe this should not impact the generalisability of the results**'.

Before external validation on the BIDMC cohort, our prediction system AIHFLevel underwent rigorous evaluation within our training dataset (Discovery cohort) and testing datasets (Replication and Meta cohorts) [Subsection3: Uniformly prognostic implication and predictive performance of AIHFLevel; Subsection4: Robustness and superior performance of AIHFLevel; Subsection5: Clinical interpretability underlying AIHFLevel; Subsection6: Prognostic stratification underlying AIHFLevel; Subsection7: Sensitivity analysis] (**revised manuscript: lines 158-335, pages 7-13**). Given that the Replication cohort and Meta cohort still belong to internal testing datasets with homogenous data patterns, ethnic backgrounds, and medical system technology platforms, conducting independent validation on a highly heterogeneous external cohort remains a priority to demonstrate the generalizability and extrapolation capabilities of the prediction system.

(1) As emphasized by one latest review, prior to the implementation of ML algorithms, external validation is imperative as prediction models may perform worse on external datasets. Notably, most of the studies done in AI are not externally validated. These limitations can potentially be addressed by performing validation on multiple independent datasets or different data modalities (i.e. datasets from various regions and hospitals) for predictive analytics. Prediction models can correspond erroneously or be fitted to peculiarities in the development dataset. As we move towards applying predictive modelling in clinical practice that aims to personalize care, existing methods or prospective proposals must be systematically evaluated

on datasets containing diverse populations from real-world clinical settings (*Artificial intelligence and heart failure: A state-of-the-art review. Eur J Heart Fail. 2023 Sep;25(9):1507-1525. PMID: 37560778.*).

(2) In general, incorporating an AI model into clinical practice requires satisfying multiple criteria, particularly the need for algorithms to be tested across multiple cohorts and to demonstrate universal applicability across various healthcare settings, systems, and geographic locations. Diverse databases may contain poorly classified or reported variables and employ different technologies and platforms, highlighting the importance of validation using independent cohorts (*Risk prediction models: II. External validation, model updating, and impact assessment. Heart. PMID: 22397946*). Moreover, the timing of endpoint event observation largely depends on the surveillance schedule, and the follow-up protocol can influence patient survival, thereby affecting the model's performance in different settings (*Database Selection and Heterogeneity-More Details, More Credibility. JAMA Oncol. PMID: 29955774*). Hence, validation across different data modalities (i.e. datasets from diverse hospital systems technology platforms, even regions, and ethnic backgrounds) for predictive analytics is essential (*Contemporary Applications of Machine Learning for Device Therapy in Heart Failure. JACC Heart Fail. PMID: 36049812*). In this study, the disparity in follow-up, attrition, detection, and selection schedules between CRCCD Center and Beth Israel Deaconess Medical BIDMC Center provided a robust test bed for assessing the AIHFLevel's generalizability (**Revised Supplementary Table 18**). Despite the challenges, AIHFLevel has been confirmed as a robust tool for risk stratification and estimation, as evidenced by rigorous assessments of various criteria, including discrimination, calibration, performance, generalization, and clinical utility in the Discovery, Replication, Meta, and External heterogeneous cohorts.

The revised content mentioned above has been incorporated into the 'Extrapolation of AIHFLevel to the heterogeneous populations' subsections of the Results section (**revised manuscript: line 337-374, page 13-14**), as well as in the 'Para 3' subsection of the Discussion section (**revised manuscript: line 449-465, page 17**).

Thank you once again for your valuable and constructive feedback. We believe your insights will greatly enhance the rigor, and credibility of our study. We sincerely hope for your approval.

Reviewer #3 (Remarks to the Author):

This work developed an effective survival prediction model by integrating diverse features from medical records as well as clinical markers using 12 machine learning methods for heart failure with renal dysfunction. They demonstrated the outperformance of the proposed model/system CPS in both internal and external clinical cohorts when compared with existing survival prediction tools. My comments as follows:

Response:

Thank you for your insightful advice. We have thoroughly re-examined and revised our manuscript. As you rightly pointed out, there were several areas requiring further clarification and enhancement, including the motivation behind the study, the process of prediction, model comparison, and figure, and legend.

In response to your valuable comments, we have modified and restructured the revised manuscript to enhance readability and comprehension. We have included additional data analyses and provided a detailed discussion to properly interpret the AIHFLevel prediction system and give it fuller clinical implications. Detailed information and raw data related to the analysis results have been added to enhance the reproducibility of the research data, and are now included in the **revised Supplementary Tables 1-20**. Each figure and Supplementary figure has been polished and restructured for clarity and readability in the **revised Figures 1-7** and **revised Supplementary Figures 1-10**. Please find the **overview of the revised manuscript and research** in the responses to Reviewer #1 in **(Remarks to the Author)**. We will clarify and address each of your suggestions in turn.

We sincerely hope that our revisions will meet with your esteemed approval!

Q1: The key standing point of the study is there is a lack of composite approaches with machine learning for survival prediction in AHF patients with RD (line 83). However, there is little that has been discussed on why standalone studies are needed here, rather than using approaches for AHF or other similar diseases (there is a wide range of prediction models), despite general challenges for survival prediction in the clinic were mentioned (lines 85-92). The key references 8 and 9 are somewhat outdated as well. So there appears to be a gap in the rationale for the motivation behind the study, specifically regarding the clinical evidence motivating this study for AHF with RD.

Response:

Thank you for your insightful inquiry. We are pleased to have the opportunity to clarify and provide the motivation and rationale for our study further. Following your suggestion, we have consulted a wide range of literature and conducted an in-depth investigation. The newly added content has been incorporated into the revised manuscript, and we hope that our revisions will receive your approval.

(1) Why we should be concerned about advanced heart failure (AHF)?

Heart failure (HF) is a life-threatening condition marking the final common pathway for many cardiac diseases. The prognosis of patients with HF has improved considerably since the publication of the first treatment trials a few decades ago. However, it remains poor, and quality of life (QOL) is also markedly reduced. (*Eur J Heart Fail. 2018 Nov. PMID: 29806100*) (*2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure, Eur Heart J, PMID: 34447992, Subsection 3.3.3 Natural history and prognosis (Page 3614)*). More and more patients with HF progress into a phase of advanced HF (AHF), characterized by persistent symptoms despite maximal therapy. (*Advanced (stage D) heart failure: a statement from the Heart Failure Society of America Guidelines Committee. J Card Fail. PMID: 25953697; Advanced Heart Failure: Epidemiology, Diagnosis, and Therapeutic Approaches. JACC Heart Fail. PMID: 32535126*). The prevalence of AHF is increasing due to the growing number of patients with HF, ageing of the population, and better treatment and survival of HF. Prognosis remains poor, with **a 1-year mortality ranging from 25% to 75%** (*JACC Heart Fail, PMID: 27395350*) (*2021 ESC Guidelines, Subsection 10.1 Epidemiology, diagnosis, and prognosis (Page 3637)*). More importantly, as European Society of Cardiology indicated that while the disease trajectory of each patient with HF is unique, there is a generalizable pattern of gradual decline, punctuated by episodes of acute deterioration leading

either to sudden death or death due to progressive HF. Thus, **anticipatory planning should start to prevent sudden death when a patient is diagnosed with AHF** (2021 ESC Guidelines, Subsection 10.2.4 Symptom control and end-of-life care (page 3644)). Additionally, **prognostic stratification** is important to identify the ideal time for referral to an appropriate center (i.e. one capable of providing AHF therapies), to properly convey expectations to patients and families, and to plan treatment and follow-up strategies (2021 ESC Guidelines, Subsection 10.1 Epidemiology, diagnosis, and prognosis (page 3637)). Importantly, patients with mild symptoms may still have a high risk of hospitalization and death.²⁰ Predicting outcome is particularly important in AHF to guide selection of cardiac transplantation and device therapies (2021 ESC Guidelines, Subsection 10.1 Epidemiology, diagnosis, and prognosis (page 3637)).

Despite many prognostic parameters and tools, predicting outcomes remains difficult and patients are often referred to AHF centers too late. In the guideline, researchers have decided to focus on the prognosis and treatment of HF, not on its prevention. (2023 Focused Update of the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2023 Oct. PMID: 37622666).

(2) Why we should be concerned about AHF combined with renal dysfunction (RD)?

Based on your kind suggestions, regarding the comorbidity of AHF and renal dysfunction, to begin with, we studied and updated a substantial amount of recent authoritative literature:

(a) Association of chronic kidney disease with abnormal cardiac mechanics and adverse outcomes in patients with heart failure and preserved ejection fraction. Eur J Heart Fail. PMID: 26635076;

(b) Associations with and prognostic impact of chronic kidney disease in heart failure with preserved, mid-range, and reduced ejection fraction. Eur J Heart Fail. PMID: 28371075;

(c) Renal dysfunction in patients with heart failure with preserved versus reduced ejection fraction: impact of the new Chronic Kidney Disease-Epidemiology Collaboration Group formula. Circ Heart Fail. PMID: 22441773;

(d) Heart Failure Increases the Risk of Adverse Renal Outcomes in Patients With Normal Kidney Function. Circ Heart Fail. PMID: 28765150;

(e) *Effect of renal function on prognosis in chronic heart failure. Am J Cardiol. PMID: 25465925;*

(f) *Clinical characteristics and prognostic influence of renal dysfunction in heart failure patients with preserved ejection fraction. Eur J Intern Med. PMID: 23830876;*

(g) *Renal function as a predictor of outcome in a broad spectrum of patients with heart failure. Circulation. PMID: 16461840;*

(h) *Advanced heart failure in adult congenital heart disease: the role of renal dysfunction in management and outcomes. Eur J Prev Cardiol. PMID: 36974357;*

(i) *Evidence-Based Medical Therapy in Patients With Heart Failure With Reduced Ejection Fraction and Chronic Kidney Disease. Circulation. PMID: 35226558;*

(j) *Kidney Function and Outcomes in Patients Hospitalized With Heart Failure. J Am Coll Cardiol. PMID: 33989713;*

(k) *Burden and Outcomes of Heart Failure Hospitalizations in Adults With Chronic Kidney Disease. J Am Coll Cardiol. PMID: 31146814;*

We found that almost all the gathered resources collectively highlight the significance of investigating this comorbidity. Herein, we aim to provide a systematic and succinct overview for your consideration.

Initially, chronic kidney disease (CKD) has consistently been recognized as one of the most prevalent comorbidities in HF (*Eur J Heart Fail, PMID: 28371075*) (*2023 Focused Update of the 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure, Eur Heart J, PMID: 37622666*), when present, carried the highest population attributable risk for all-cause mortality (ACM) and AHF hospitalization among all comorbidities. (*Circulation, PMID: 35226558*) (*Int J Cardiol, PMID: 28528982*) (*JACC Heart Fail, PMID: 30606484*). Renal dysfunction (RD) is now regarded as a major independent determinant of increased mortality and morbidity in AHF (*Prog Cardiovasc Dis. PMID: 31377223*) (*Nat Rev Nephrol. PMID: 27573728*) (*J Card Fail. PMID: 20447572*) (*Am Heart J. PMID: 22424011*).

Kidney dysfunction frequently coexists with chronic HF, and the presence of both is associated with worse clinical outcomes than either condition alone (*JAMA Cardiol. PMID: 28002548*). HF and RD share common risk factors, and more importantly, their complex interplay accelerated disease progression, driven by neurohormonal and inflammatory

activation, elevated venous pressure, and hypoperfusion (*Eur Heart J*, PMID: 34447992). In clinical practice, patients misclassified as having renal dysfunction may be exposed to potential harms because such patients often are treated with lower doses of drugs, and diagnostic tests using contrast media are avoided. (*Eur J Heart Fail*, PMID: 26635076). As noted in one recent study, despite facing elevated risks of mortality, patients with comorbid AHF and RD are not optimally treated with evidence-based medical therapies, even at levels of eGFR where such therapies would not be contraindicated by kidney dysfunction. Further efforts are required to mitigate risk in comorbid HF and kidney disease (*J Am Coll Cardiol*, PMID: 33989713). The 2021 Guideline also indicated that there is little direct evidence to support any recommendations for treatment of HF patients with severe CKD as to date (2021 ESC Guidelines, Subection 13.6 Kidney dysfunction (page 3669)).

While emerging novel biomarkers provided some post-discharge prognostic value, they are not specific enough for AHF or RD cases, constraining their clinical utility (*JACC Heart Fail*, PMID: 36114137) (*Ann Intern Med*, PMID: 31842204) (*JAMA Cardiol*, PMID: 30140899) (*J Card Fail*, PMID: 30965102) (*J Pers Med*, PMID: 35743683). The traditional biomarker natriuretic peptides may guide further cardiac investigation. However, it should be noted that there are many causes of an elevated NP—both cardiovascular and non-cardiovascular—that might reduce their diagnostic accuracy. These causes include acute or chronic kidney disease (2021 ESC Guidelines, Subsection 4.2 Natriuretic peptides (page 3617)). Additionally, the efficacy of management and prognostication guidance using serial natriuretic peptide measurements remained unestablished (*Eur J Heart Fail*, PMID: 37098791). Heterogeneity within HF and RD populations reduced the efficacy of a single biomarker (*Circulation*, PMID: 29792299).

Therefore, comprehensive survival predictions should be prioritized in this high-risk population for clear prognostication and targeted interventions (*Circ Heart Fail*, PMID: 28765150; *J Am Coll Cardiol*, PMID: 33989713; *Eur Heart J*, PMID: 36691956).

(3) Obstacles in HF AI Modeling: Our Goals for Improvement:

(I) The high dimensionality and interactivity of clinical data rendered hypothesis-driven traditional statistical methods less effective. Currently, the predictive capability of conventional prognostic models for HF was generally limited (*Contemporary Applications of Machine Learning for Device Therapy in Heart Failure. JACC Heart Fail. PMID: 36049812*) (*Clinical applications of machine learning in the diagnosis, classification, and prediction of heart failure. Am Heart J. PMID: 32905873*).

(II) Artificial intelligence (AI) is increasingly employed to establish prognostic tools for predicting death, readmission or composite endpoints; however, no algorithm can perfectly assess HF outcomes (*A cardiologist's guide to machine learning in cardiovascular disease prognosis prediction. Basic Res Cardiol. 2023 Mar. PMID: 36939941*).

(III) Despite many options available, researchers' preferences and knowledge limitations can lead to suboptimal use of modeling algorithms, resulting in reduced predictive power (*Predictive models for clinical decision making: Deep dives in practical machine learning. J Intern Med. 2022 Aug; PMID: 35426190*).

(IV) Despite many options available, researchers' preferences and knowledge limitations can lead to suboptimal use of modeling algorithms, resulting in reduced predictive power (*Predictive models for clinical decision making: Deep dives in practical machine learning. J Intern Med. 2022 Aug; PMID: 35426190*). In clinical practice, it is essential to comprehensively consider various research objectives, implementation conditions, and expected outcomes, and make appropriate choices based on the currently known rules (*Cardiovascular complications in a diabetes prediction model using machine learning: a systematic review. Cardiovasc Diabetol. 2023 Jan PMID: 36658644*).

(V) Additionally, current risk models, often burdened with redundant or costly predictors, prove impractical for daily clinical practice and inaccessible in primary care. Due to large number of predictor variables used in the study, the generalizability of high-dimensional ML models can still be constrained (*Artificial intelligence and prediction of cardiometabolic disease: Systematic review of model performance and potential benefits in indigenous populations. Artif Intell Med. 2023 May. PMID: 37100506*).

(VI) The real-world application of predictive models requires balancing accuracy, interpretability, and complexity. Despite the potency, AI's 'black-box' nature posed challenges in providing clear interpretations and transparency to clinicians, a key factor limiting clinical implementation of many HF prognostic models. This is because, in addition to prognostic endpoints, both physicians and patients are concerned about the substantial risks associated

with the various predictors, which can assist physicians in their decision-making (*Scalable and accurate deep learning with electronic health records. NPJ Digit Med. 2018 May. PMID: 31304302*) (*Scalable and accurate deep learning with electronic health records. NPJ Digit Med. 2018 May. PMID: 31304302*) (*Steps to avoid overuse and misuse of machine learning in clinical research. Nat Med. 2022 Oct. PMID: 36097217*) (*Contemporary Applications of Machine Learning for Device Therapy in Heart Failure. JACC Heart Fail. 2022 Sep. PMID: 36049812*).

(VII) Moreover, the lack of extensive external validation and calibration in many HF prognostic models restricted their generalizability across diverse clinical settings and populations (*Machine Learning and Social Determinants of Health-An Opportunity to Move Beyond Race for Inpatient Risk Prediction in Patients With Heart Failure. JAMA Cardiol. 2022 Aug. PMID: 35793074*).

(4) AIHFLevel: A Web-based System for Clinician Use:

In this study, we aimed to develop and validate an explainable predictive system for survival assessment of patients with AHF and RD in multicenter retrospective longitudinal cohorts. Our AI hybrid modeling framework maximized the use of various algorithms, reducing the impact of researcher bias. We also integrated the system with clinical interpretability and prognostic stratification, optimizing patient management strategies. This system (AIHFLevel) has been translated into a convenient web application to facilitate its utility for clinicians, available at 'www.hf-ai-survival.com'.

The revised content mentioned above has been incorporated into the Introduction section (revised manuscript: line 33-88, page 3-4), and the 'Discussion' section (revised manuscript: line 410-556, page 16-23).

Once again, we sincerely appreciate your insightful suggestions, which we're confident will greatly improve the robustness, reliability of our research. We genuinely hope for your support and endorsement!

Q2: The process of CPS system development (lines 157-203) isn't quite clear to me. A graphic illustration would be very helpful, e.g. how features are initially selected as well as for each ML model, how were they combined, which set of data was used at each training/validation step. Given this is one of the key contributions of the study, a more organized and clear presentation is essential in the Method.

Response:

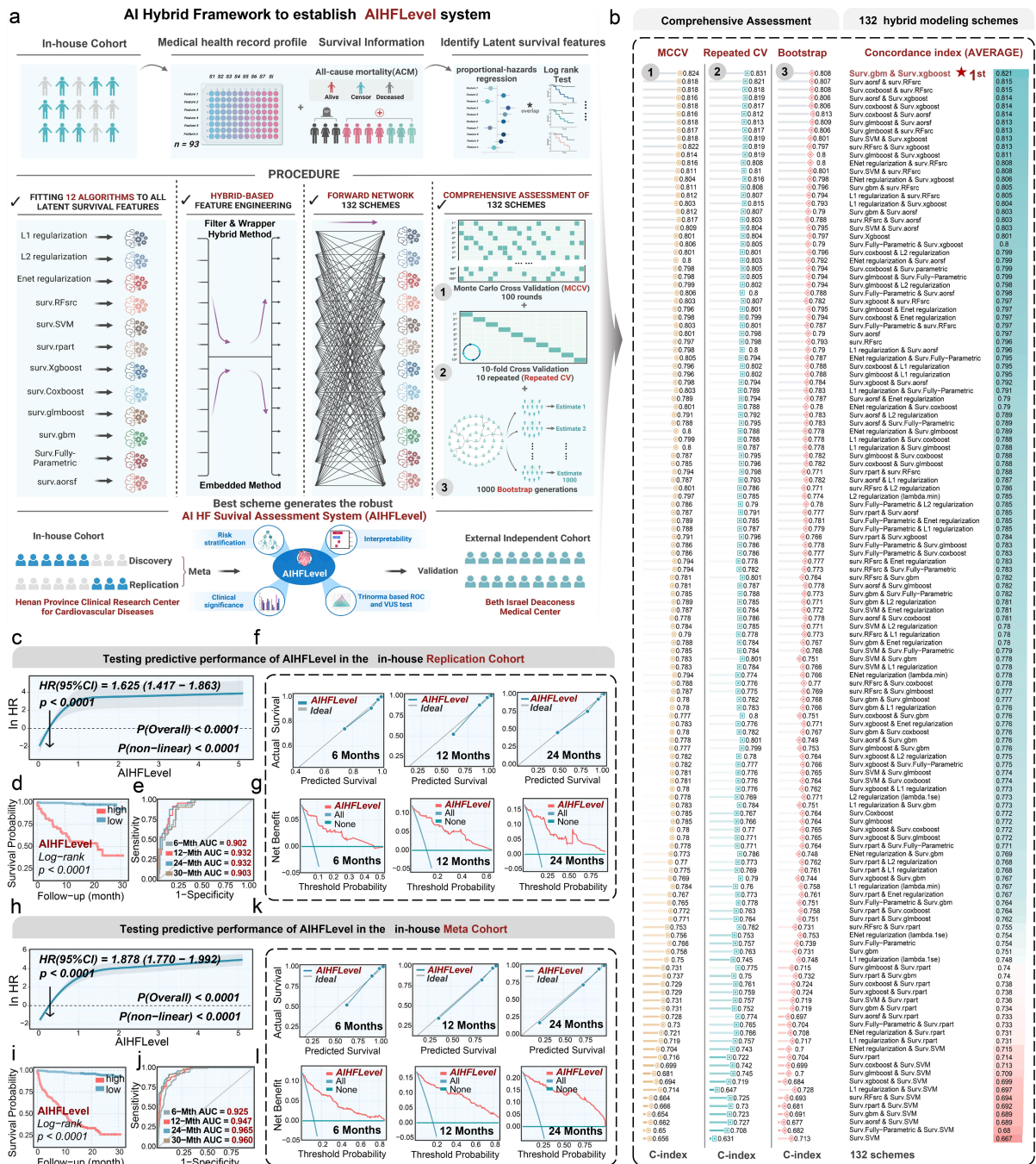
We are grateful for your thoughtful feedback and attention to this matter. We apologize for any confusion caused by the previous manuscript. We have enhanced the revised manuscript with more detailed textual descriptions and illustrative figures. We sincerely hope these revisions meet your expectations.

(1): Why we committed to develop an AI hybrid modeling framework? (Revised manuscript lines 782-794, page 28-29).

There are many modeling algorithms to analyze data and assess outcomes currently. Predictions provided by different algorithms may vary depending on the characteristics of the dataset. It remained challenging to identify the appropriate learner tailored to the application's requirements. Despite many options available, researchers' preferences and knowledge limitations can lead to suboptimal use of modeling algorithms, resulting in reduced predictive power (*J Intern Med*, PMID: 35426190). Algorithm selection should be based on various objectives, conditions, aligned with known rules (*Cardiovasc Diabetol*. PMID: 36658644).

However, some researchers have suggested that the use of hybrid learning models, e.g. ensemble-based approaches that is, a combination of two models, and enhancing existing learning models can be helpful. Mienye and colleagues provided evidence that ensemble-based learning may perform better than the single model in the prediction of HF events (*Informatics in Medicine Unlocked*, <https://doi.org/10.1016/j.imu.2020.100402>). Ensemble modeling operates by employing a collection of diverse and independent models to make predictions, thereby reducing the generalization error and improving the reliability of the predictive outcomes (*AMIA Annu Symp Proc*, PMID: 30815108).

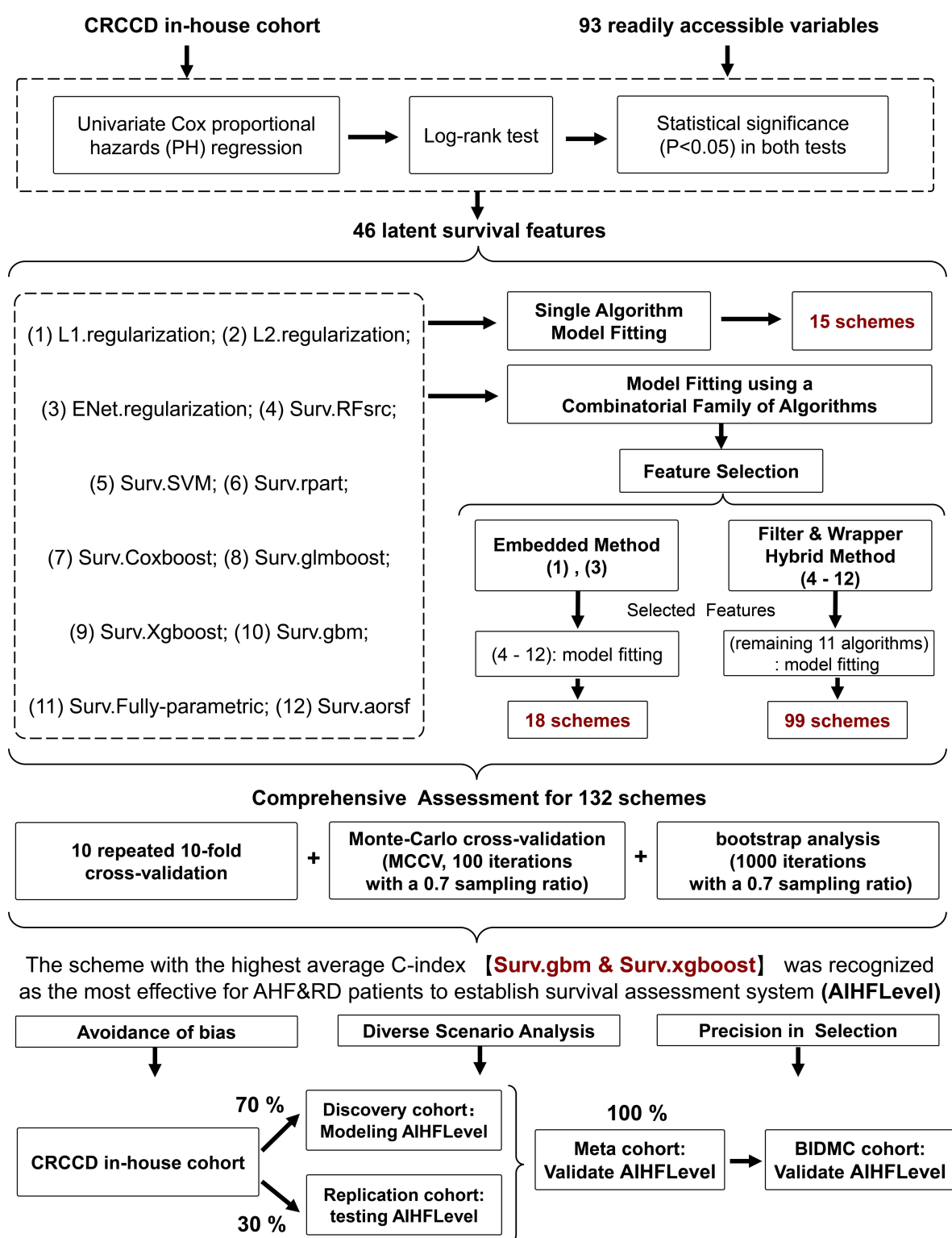
The graphic illustration was presented in (Revised Fig. 2a, Revised Supplementary Fig. 9)



Revised Fig. 2: Consistent predictive performance and prognostic value of AIHFLevel.

a AI modeling hybrid framework overview.

b Schematic illustration of optimal scheme identification. The diagram delineates the comprehensive evaluation process of 132 distinct modeling schemes through an array of lollipop plots, including 10 repeated 10-fold cross-validation, 100 iterations of Monte-Carlo cross-validation (MCCV) with a 70% sampling ratio, and a thorough bootstrap analysis comprising 1000 iterations. This process pinpointed the combined scheme Surv.gbm&Surv.Xgboost, enhanced by the Filter & Wrapper Hybrid Method, yielded the highest average C-index, demonstrating effective prognostic precision for AHF&RD cases.



Supplementary Figure 9. Graphic illustration of AI hybrid modeling framework generated survival assessment system (AIHFLevel).

(2): How dose it work? (Revised manuscript line 796-886, pages 29-32).

(I) Dataset: The CRCCD in-house cohort was used to evaluate the viability of various algorithms and identify the optimal modeling schemes, while the BIDMC external cohort served to validate the system's generalizability.

ACM serves as an outcome to train our system.

(II) Variable preprocessing: We incorporated 93 readily accessible variables, as detailed earlier **(Revised manuscript page 664-705)**. In the CRCCD cohort, univariate Cox proportional hazards (PH) regression and Log-rank test were conducted. 46 variables were identified as latent survival features, demonstrating statistical significance ($P < 0.05$) in both tests.

(III) 12 modeling algorithms: L1-regularized PH regression (L1.regularization), L2-regularized PH regression (L2.regularization), Elastic-Net-regularized PH regression (ENet.regularization), Survival random forest SRC learner (Surv.RFsrc), Survival support vector machine (Surv.SVM), survival tree (Surv.rpart), Cox model with likelihood-based boosting (Surv.Coxboost), Boosted generalized linear survival learner (Surv.glmboost), Extreme gradient boosting survival learner (Surv.Xgboost), Survival gradient boosting machine learner (Surv.gbm), Survival fully parametric learner (Surv.Fully-parametric), Accelerated oblique random survival forest learner (Surv.aorsf). Details of their hyperparameters are presented in **(Revised Supplementary Table 10)**.

Supplementary Table 10. Hyperparameters for the 12 modeling algorithms

L1-regularized PH regression (L1.regularization)	Alignment = lambda; alpha = 1; big = 9.9e+35; devmax = 0.999; eps = 1e-06; epsnr = 1e-08; exmx = 250; fdev = 1e-05; grouped = TRUE; intercept = TRUE; keep = TRUE; maxit = 100000; mnlam = 5; mxit = 100; mxitnr = 25; nfolds = 10; nlambdas = 100; parallel = FALSE; pmin = 1e-09; prec = 1e-10; predict.gamma = gamma.1se; relax = FALSE; s = (lambda.1se, lambda.min); standardize = TRUE; standardize.response = FALSE; thresh = 1e-07; trace.it = 0; type.logistic = Newton; type.measure = C; type.multinomial = ungrouped; upper.limits = Inf
L2-regularized PH regression (L1.regularization)	Alignment = lambda; alpha = 0; big = 9.9e+35; devmax = 0.999; eps = 1e-06; epsnr = 1e-08; exmx = 250; fdev = 1e-05; grouped = TRUE; intercept = TRUE; keep = TRUE; maxit = 100000; mnlam = 5; mxit = 100; mxitnr = 25; nfolds = 10; nlambdas = 100; parallel = FALSE; pmin = 1e-09; prec = 1e-10; predict.gamma = gamma.1se; relax = FALSE; s = (lambda.1se, lambda.min); standardize = TRUE; standardize.response = FALSE; thresh = 1e-07; trace.it = 0; type.logistic = Newton; type.measure = C; type.multinomial = ungrouped; upper.limits = Inf
ENet-regularized	Alignment = lambda; alpha = 0.3285714; big = 9.9e+35; devmax = 0.999;

PH regression (L1.regularization)	eps = 1e-06; epsnr = 1e-08; exmx = 250; fdev = 1e-05; grouped = TRUE; intercept = TRUE; keep = TRUE; maxit = 100000; mnlam = 5; mxit = 100; mxitnr = 25; nfolds = 10; nlambda = 100; parallel = FALSE; pmin = 1e-09; prec = 1e-10; predict.gamma = gamma.1se; relax = FALSE; s = (lambda.1se, lambda.min); standardize = TRUE; standardize.response = FALSE; thresh = 1e-07; trace.it = 0; type.logistic = Newton; type.measure = C; type.multinomial = ungrouped; upper.limits = Inf
Survival random forest SRC learner (Surv.RFsrc)	ntree = 1000; nodesize = 15; splitrule = logrank; nsplit = 10; importance = FALSE; block.size = 10; bootstrap = by.root; samptype = swor; membership = FALSE; na.action = na.omit; nimpute = 1; proximity = FALSE; distance = FALSE; forest.wt = FALSE; forest = TRUE; var.used = by.tree; split.depth = by.tree; do.trace = TRUE; statistics = TRUE; estimator = nelson
Survival support vector machine (Surv.SVM)	type = regression; opt.meth = quadprog; kernel = lin_kernel; sgf.sv = 5; sigf = 7; maxiter = 20; margin = 0.05; bound = 10; eig.tol = 1e-06; conv.tol = 1e-07; posd.tol = 1e-08
Survival tree (Surv.rpart)	parms = 1; minbucket = 3; minsplit = 20; cp = 0.01; maxcompete = 4; maxsurrogate = 5; maxdepth = 30; usesurrogate = 2; surrogatestyle = 0; xval = 10; cost = 3
Cox model with likelihood-based boosting (Surv.Coxboost)	standardize = TRUE; stepno = 100; criterion = pscore; stepsize.factor = 1; sf.scheme = sigmoid; x.is.01 = FALSE; return.score = TRUE; trace = FALSE
Boosted generalized linear survival learner (Surv.glmboost)	family = coxph; nuirange = c , 0 , 100; center = TRUE; mstop = 1000; nu = 0.1; risk = inbag; stopintern = FALSE; trace = FALSE; sigma = 0.1; ipcw = 1; na.action = :: , stats , na.omit
Extreme gradient boosting survival learner (Surv.Xgboost)	aft_loss_distribution = normal; alpha = 0; base_score = 0.5; booster = gbtrees; colsample_bylevel = 1; colsample_bynode = 1; colsample_bytree = 1; disable_default_eval_metric = FALSE; eta = 0.3; feature_selector = cyclic; gamma = 0; grow_policy = depthwise; lambda = 1; lambda_bias = 0; max_bin = 256; max_delta_step = 0; max_depth = 6; max_leaves = 0; min_child_weight = 1; monotone_constraints = 0; normalize_type = tree; nthread = 1; num_parallel_tree = 1; objective = survival:cox; one_drop = FALSE; predictor = cpu_predictor; print_every_n = 1; process_type = default; rate_drop = 0; refresh_leaf = TRUE; sample_type = uniform; sampling_method = uniform; scale_pos_weight = 1; seed_per_iteration = FALSE; skip_drop = 0; strict_shape = FALSE; subsample = 1; top_k = 0; tree_method = auto; tweedie_variance_power = 1.5; verbose = 1
Survival gradient boosting machine learner (Surv.gbm)	distribution = 1; n.trees = 1000; cv.folds = 10; interaction.depth = 1; n.minobsinnode = 10; shrinkage = 0.001; bag.fraction = 0.5; train.fraction = 0.7; keep.data = FALSE; verbose = FALSE; single.tree = FALSE
Survival fully parametric learner (Surv.Fully-parametric)	type = atf; dist = Weibull; scale = 0; maxiter = 30; rel.tolerance = 1e-09; toler.chol = 1e-10; debug = 0; outer.max = 10; robust = FALSE; score = FALSE
Accelerated	n_tree = 500; n_split = 5; n_retry = 3; control_type = fast;

oblique survival learner (Surv.aorsf)	random forest	control_fast_do_scale = TRUE; control_fast_method = Efron; control_cph_method = Efron; control_cph_eps = 1e-09; control_cph_iter_max = 20; control_net_alpha = 0.5; control_net_df_target = NULL; leaf_min_events = 1; leaf_min_obs = 5; split_min_events = 5; split_min_obs = 10; split_min_stat = 3.841459; oobag_pred_type = surv; importance = anova; attach_data = TRUE; verbose_progress = FALSE; na_action = fail
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(IV) 132 modeling schemes:

(a) Single Algorithm Model Fitting:

12 algorithms were performed on 46 latent survival features to fit models, generating 15 schemes: L1.regularization (lambda.min), L1.regularization (lambda.1se), L2.regularization (lambda.min), L2.regularization (lambda.1se), ENet.regularization (lambda.min), ENet.regularization (lambda.1se), Surv.RFsrc, Surv.SVM, Surv.rpart, Surv.Coxboost, Surv.glmboost, Surv.Xgboost, Surv.gbm, Surv.Fully-parametric, Surv.aorsf. For regularization algorithms (L1, L2, and ENet), we utilized 'lambda.min' and 'lambda.1se' as λ regularization parameters to expand the range of possible schemes.

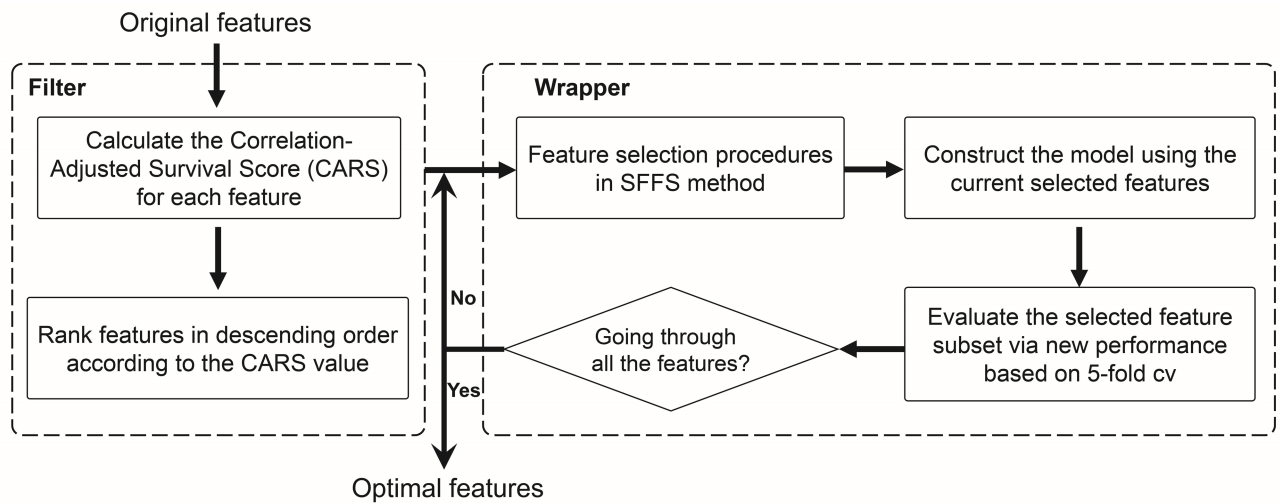
(b) Model Fitting using a Combinatorial Family of Algorithms:

Feature selection on 46 latent survival features was initially executed with each algorithm, followed by fitting the selected features with the remaining algorithms.

Embedded Method: For L1.regularization and ENet.regularization, we directly leveraged their built-in selection mechanisms based on L1 and elastic-net penalties. 5-fold cross-validation served as the resampling strategy for evaluating feature subset performance. The other 9 algorithms, excluding regularization methods, were then applied to fit models on the selected features, generating 18 schemes.

Filter & Wrapper Hybrid Method: This method combined the rapid training capability of the external technique Filter (feature ranking) and the intensive computation of the Wrapper (using the model as a black box to evaluate feature subset according to their predictive power) (*A survey on feature selection methods, Computers & Electrical Engineering, <https://doi.org/10.1016/j.compeleceng.2013.11.024>*). Apart from the regularization methods, each of the remaining 9 algorithms executed the following steps: First, the filter method was adopted to rank the features in descending order according to the Correlation-Adjusted Survival Score (CARS). Next, sequential forward floating selection (SFFS) as a wrapper method was used to select features. Features were added in feature sets in sequence, and feature retention

was based on the performance of each algorithm at every step. The 5-fold cross-validation assisted the performance assessment. If the performance with a new feature set did not increase, the new feature was removed from the feature set. (**Revised Supplementary Fig. 4**) illustrated the main procedure of the Filter & Wrapper Hybrid Method. Finally, the remaining 11 algorithms were used to fit models on the selected features, generating 99 schemes.



Supplementary Figure 10. Filter & Wrapper Hybrid Method.

The 9 algorithms, except for regularization methods, perform the following steps in sequence for feature selection: (Surv.RFsrc, Surv.SVM, Surv.rpart, Surv.Coxboost, Surv.glmboost, Surv.Xgboost, Surv.gbm, Surv.Fully-parametric, Surv.aorsf)

Step 1. Calculate the Correlation-Adjusted Survival Score (CARS): for each algorithm, the CARS was computed for each feature, and the features were ranked in descending order.

Step 2. Build a model: CRCCD in-house cohort was randomly divided into five groups using a 5-fold cross-validation method. Each time, four groups were selected as the training subset, and the remaining group was used as the test subset. The final performance was the average result of the 5 testing assessments. For each training subset, the sorted features were added to the feature set in turn; the feature set was initially empty. The model was constructed using the selected features.

Step 3. Apply search strategy: according to the sequential forward floating selection (SFFS) strategy and the performance of the model, if the new performance was not improved, the newly added feature was removed from the feature subset. Otherwise, the feature was retained.

Step 4. Steps 3 and 4 were repeated until all features were selected.

Finally, the remaining 11 algorithms were used to fit models on the selected features, generating 99 schemes.

(V) Comprehensive Assessment:

In the CRCCD cohort, we implemented 132 modeling schemes, each generating a separate model. The evaluation of these models' effectiveness and relevance was conducted via a rigorous comprehensive assessment strategy. This included:

- 1. 10 repeated 10-fold cross-validation,**
- 2. Monte-Carlo cross-validation (MCCV, 100 iterations with a 0.7 sampling ratio),**
- 3. bootstrap analysis (1000 iterations with a 0.7 sampling ratio).**

The concordance index (C-index), calculated across three strategies, served as the pivotal performance metric. The modeling scheme with the highest average C-index was recognized as the most effective for AHF&RD patients.

This consensus assessment strategy offered **advantages**:

(a) **Comprehensive Coverage:** Each assessment method has both advantages and disadvantages. Integrating multifaceted validation techniques ensured a more thorough and resilient evaluation. This multifaceted approach countered the risks of overfitting, bias inherent in single-method validation and selection bias by researchers.

(b) **Diverse Scenario Analysis:** Using MCCV and bootstrap analysis allowed for an extensive examination of the models' performance across diverse datasets and hypothetical clinical scenarios, thus ensuring that the model is not overly tailored to a specific dataset or condition. This enhances the model's generalizability, making it more applicable and reliable in different clinical settings and populations.

(c) **Precision in Model Selection:** Applying the average C-index across diverse validation strategies refined the selection of the most effective model for AHF&RD, ensuring a robust choice backed by consensus performance metrics.

(d) **Reduction of Variability:** The repeated application of validation techniques, such as 10-fold cross-validation and bootstrap analysis, helps to reduce the variability in the model's performance estimates. This leads to more stable and reliable model evaluations, minimizing the impact of random fluctuations in the data.

(VI) Validation and Extrapolation: In the CRCCD in-house cohort, we divided the participants into a Discovery cohort (70%, n = 498) for model fitting using the optimal scheme, which led to the modeling of our prediction system AIHFLevel. A Replication cohort (30%, n = 214) was then employed for the initial validation of AIHFLevel. Further validation and generalization processes were conducted on the complete CRCCD Meta cohort and the external BIDMC cohort. External validation across diverse data modalities, such as patients from BIDMC with varying hospital systems, technological platforms, regional, and ethnic backgrounds, is crucial for predictive analytics. This approach can effectively address the issue of model overfitting on internal data, ensuring broader applicability and generalizability of the predictive system.

(3): The results: (Revised manuscript line 130-156, pages 6-7).

Survival assessment system AIHFLevel

The workflow of our AI hybrid framework was succinctly illustrated in **(Revised Fig. 2a)**, with detailed elaboration provided in **(Revised Supplementary Fig. 9)**. We commenced with 93 candidate predictors derived from electronic health records (EHR), narrowing them down through univariate Cox proportional hazards regression and Log-rank tests to 50 and 63 significant predictors, respectively **(Revised Supplementary Tables 6 and 7, Fig. 1d, Revised Supplementary Figs. 1 and 2)**. A core set of 46 predictors, showing consistent significance in both tests, were defined as latent survival features for inclusion in the modeling framework. 12 AI algorithms were performed on 46 latent survival features to fit models, yielding 132 distinct modeling schemes **(Revised Fig. 2b)**. Each scheme's performance was rigorously evaluated using a tripartite strategy: 10 repeated 10-fold cross-validation, Monte-Carlo cross-validation (MCCV, 100 iterations with a 0.7 sampling ratio), and bootstrap analysis (1000 iterations with a 0.7 sampling ratio) **(Revised Supplementary Table 8)**. This evaluation process identified the integration of Surv.gbm and Surv.Xgboost, enhanced by the Filter & Wrapper Hybrid Method, as the optimal modeling scheme for populations with AHF and RD. This scheme achieved the highest average C-index of 0.821, demonstrating superior discriminative power over alternative models **(Revised Fig. 2b)**.

Consequently, the AIHFLevel system was developed using the Discovery cohort, guided by this optimal scheme. AIHFLevel utilized a set of 12 readily accessible predictors: age (year), arrhythmia comorbidity (Yes or No), CAD comorbidity (Yes or No), CKD stage (I, II, III, IV, V), lymphocyte percentage (%), mean corpuscular hemoglobin concentration (MCHC, g/L), eGFR (ml/(min/1.73 m³)), serum creatinine (Cr, μmol/L), serum total bilirubin (TBIL, μmol/L), serum cardiac troponin I (CTnI, ng/ml), left ventricular ejection fraction (EF, %), stroke volume (SV, %) (**Revised Supplementary Table 9**).

We trust that these presentations and revisions align with your expectations and enhance the study's rigor. We deeply appreciate your valuable and professional guidance!

Q3: When comparing the performance of CPS with other published scores (lines 311-320), it's not clear whether the published scores/models from the referenced study or scores/models re-trained on the training data of the current study using the approach from the referenced study were applied here. If it's the former, is there any concern of bias in the clinic/cohort setting when applying these existing scores/models? Also, it's not clear to me whether the outperformance of CPS is because of the addition of the new input features (the 48 selected markers in CPS versus the input markers/features used in scores compared with; or because of the additional features from hospital records versus clinical markers only), or the modelling of the complex relationships between these features? Would be good to compare the markers/features input into each of these models, as well as the statistical/modelling approach used.

Response:

Thank you for your insightful query. We apologize for any confusion caused by the previous manuscript. We appreciate the opportunity to clarify such comparing in our study. We have enhanced the revised manuscript with more detailed textual descriptions and illustrative figures. We sincerely hope these revisions meet your expectations!

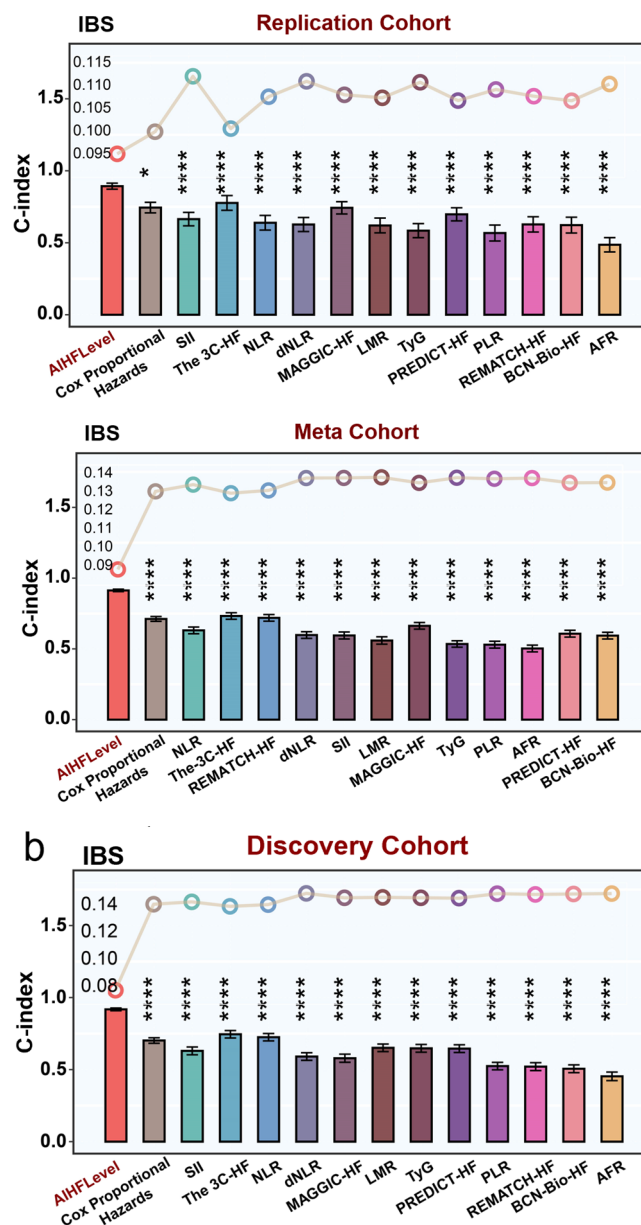
Response to the question: 'When comparing the performance of CPS with other published scores (lines 311-320), it's not clear whether the published scores/models from the referenced study or scores/models re-trained on the training data of the current study using the approach from the referenced study were applied here.'

In our previous manuscript, we used the published scores/models from the referenced study for performance comparison without retraining them. We further clarified this in the revised manuscript:

Given the development of numerous objective risk markers and composite prognostic scores for heart failure, we expanded our analysis to include a variety of published risk markers and models, including systemic immune-inflammation index (SII), neutrophil-lymphocyte ratio (NLR), neutrophils to leukocyte-neutrophil count (dNLR), lymphocyte-monocyte ratio (LMR), platelet-lymphocyte ratio (PLR), albumin-to-fibrinogen ratio (AFR), triglyceride -

glucose (TyG)-index, MAGGIC-HF, PREDICT-HF, BCN Bio-HF, REMATCH-HF, and 3C-HF score (*Eur Heart J. PMID: 23095984*) (*JAMA Cardiol. PMID: 31995119*) (*PMID: 24454874*) (*PMID: 17638928*) (*PMID: 22130225*).

Such prognostic model consisted of several predictors multiplied by corresponding coefficients to fit a risk score. To begin with, we directly computed the risk scores for each model based on their predefined features and coefficients as outlined in their original publications, and tested their predictive performance using C-index and IBS. AIHFLevel exhibited superior accuracy compared to other models (**Revised Fig. 3c, Revised Supplementary Fig. 4b, Revised Supplementary Table 11**). We observed that while some model might perform well within their original dataset, they faltered across other cohorts, likely due to a lack of generalizability and potential overfitting.



Response to the question: ‘If it’s the former, is there any concern of bias in the clinic/cohort setting when applying these existing scores/models? Also, it’s not clear to me whether the outperformance of CPS is because of the addition of the new input features (the 48 selected markers in CPS versus the input markers/features used in scores compared with; or because of the additional features from hospital records versus clinical markers only), or the modelling of the complex relationships between these features? Would be good to compare the markers/features input into each of these models, as well as the statistical/modelling approach used’

Thank you for your insightful comments! Regarding prognostic models, I'd like to share my understanding:

(1) Current prognostic models, often burdened with redundant or costly predictors:

Firstly, we'd like to clarify that in our study, we have already collected a large number of clinical variables, including 93 clinical indicators such as basic personal information, comorbidity status, blood routine, coagulation function, liver and kidney function, cardiac assessment, echocardiography results, and medication history. However, when we intended to collect more models for a comparison with our model AIHFLevel, we found that some predictors used in some other prediction models are not commonly used in daily clinical practice or not every heart failure patient would undergo those tests. ‘Expensive’ models such as the heart failure survival score (HFSS), Munich score, CORONA-HF, EFFECT, etc., were not included in our comparison due to their predictors containing unconventional clinical indicators, making it impossible for us to calculate the risk scores. Ultimately, we included SII, NLR, dNLR, LMR, PLR, AFR, TyG, MAGGIC-HF, PREDICT-HF, BCN Bio-HF, REMATCH-HF, and 3C-HF score for comparison.

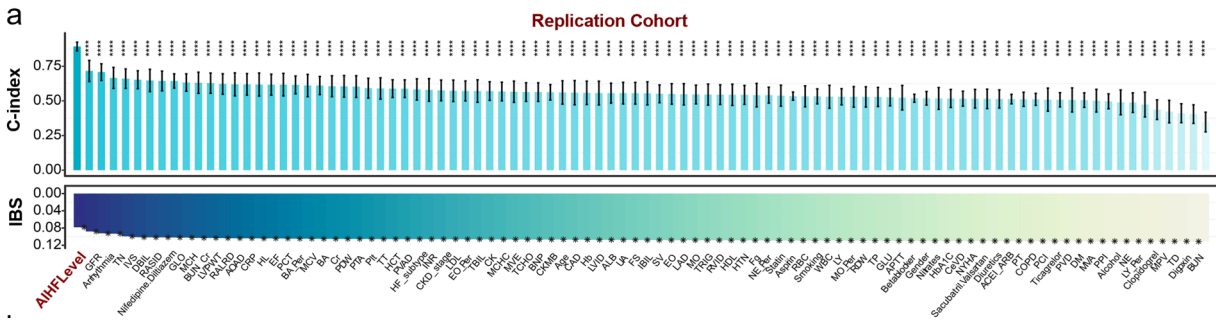
This is exactly what was mentioned in **Question 1** earlier; current prognostic models are often burdened by redundant or expensive predictors, making them impractical for routine clinical practice, especially in primary healthcare institutions. Due to large number of predictor variables used in the study, the generalizability of high-dimensional ML models can still be constrained. The application of predictive models in the real world needs to balance accuracy, interpretability, and complexity (*Artificial intelligence and prediction of cardiometabolic disease: Systematic review of model performance and potential benefits in indigenous populations. Artif Intell Med. 2023 May. PMID: 37100506*).

Our predictive system AIHFLevel just used a set of 12 readily accessible and routine clinical variables: age, arrhythmia comorbidity, CAD comorbidity, CKD stage, lymphocyte percentage, mean corpuscular hemoglobin concentration (MCHC), eGFR, serum creatinine (Cr), serum total bilirubin (TBIL), serum cardiac troponin I (CTnI), left ventricular ejection fraction (EF), and stroke volume (SV). **This is the first point where our model surpasses others.**

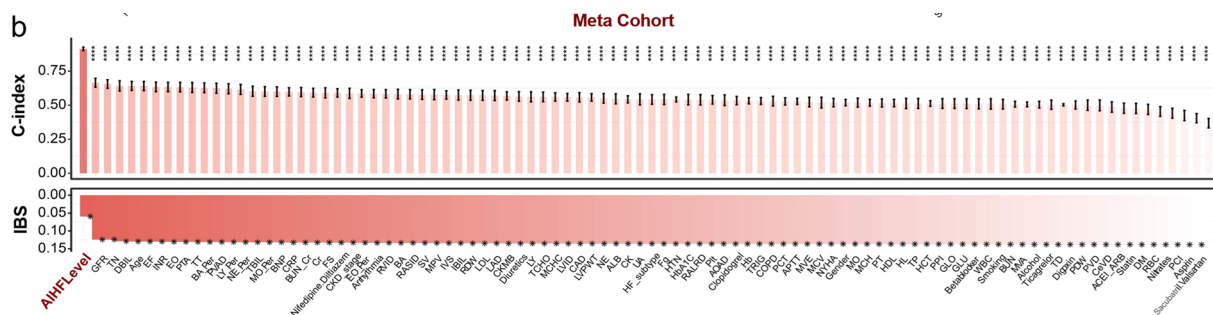
(2) Comparison with the markers/features input into each of these models:

Following your kind suggestion, **"Would be good to compare the markers/features input into each of these models"** we conducted a preliminary comparison of AIHFLevel with all the clinical variables we collected (which already include the markers/features input into each of these models) before formally comparing the models.

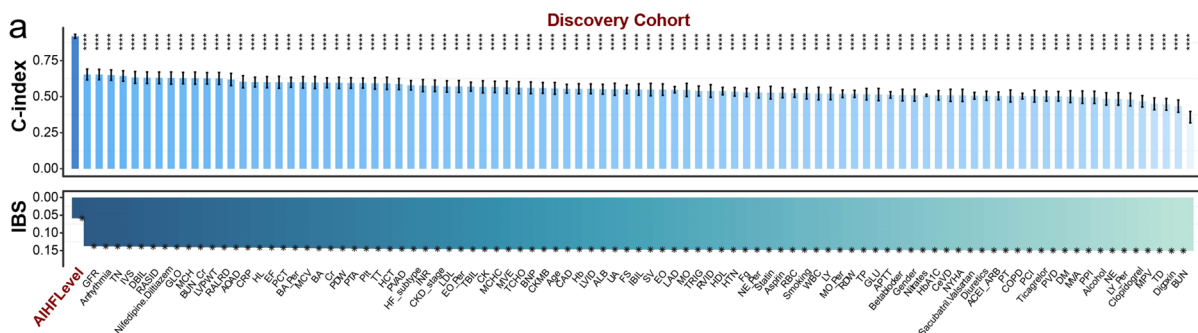
In clinical practice, prognostication and stratification for management traditionally relied on a range of clinicopathological characteristics, such as serum cardiac troponin levels, natriuretic peptides, pharmacotherapy, renal function, age, and comorbidity status.¹ To assess their predictive efficacy, we evaluated these readily accessible conventional clinical traits derived from the EHR system. Our findings revealed that AIHFLevel consistently outperformed in predictive accuracy. This superiority was quantitatively confirmed through C-index and Integrated Brier Score (IBS) across Discovery, Replication, and Meta cohorts, suggesting AIHFLevel's potential for integration into clinical workflows (**Revised Fig. 3a, b, Revised Supplementary Fig. 4a, and Revised Supplementary Table 10**).



Revised Fig. 3a



Revised Fig. 3b



Revised Supplementary Fig. 4a

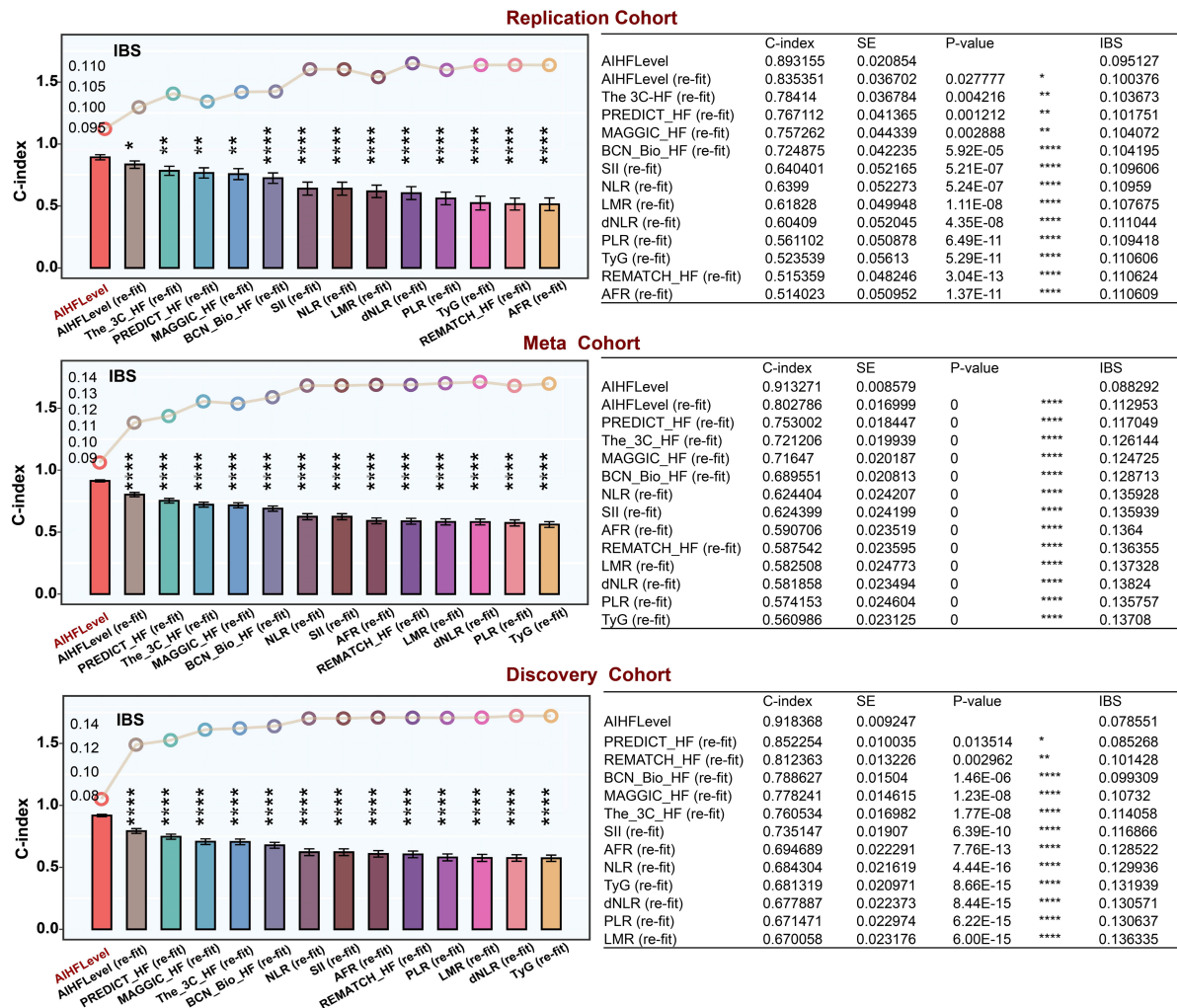
(3) In-depth analysis to further validate AIHFLevel's robustness:

As you rightly pointed out, "If it's the former, is there any concern of bias in the clinic/cohort setting when applying these existing scores/models? Would be good to compare the markers/features input into each of these models, as well as the statistical/modelling approach used" we completely agree with your perspective. For this reason, we further conducted the following two analyses:

(a) Building upon this preliminary analysis (compute the risk scores for each model based on their predefined features and coefficients as outlined in their original publications), we advanced to a secondary, in-depth evaluation to underscore the AIHFLevel's robustness and superior predictive capacity. This involved a refitting process, where AIHFLevel and other models were retrained using only their respective predictors—without incorporating their original coefficients—in a unified proportional hazards regression framework applied to the Discovery cohort. This methodological refinement aimed at facilitating an equitable, standardized comparison, thus validating the robustness of AIHFLevel in a more stringent evaluative setting. The outcomes of this recalibration underscored AIHFLevel's optimal performance, surpassing that of comparator models across the Discovery, Replication, and Meta cohorts (**Revised Supplementary Fig. 5**). This iterative validation reaffirmed

AIHFLevel's superiority over conventional models, and illuminated the resilience of its predictive capacity, demonstrating a potent tool in prognostic assessment.

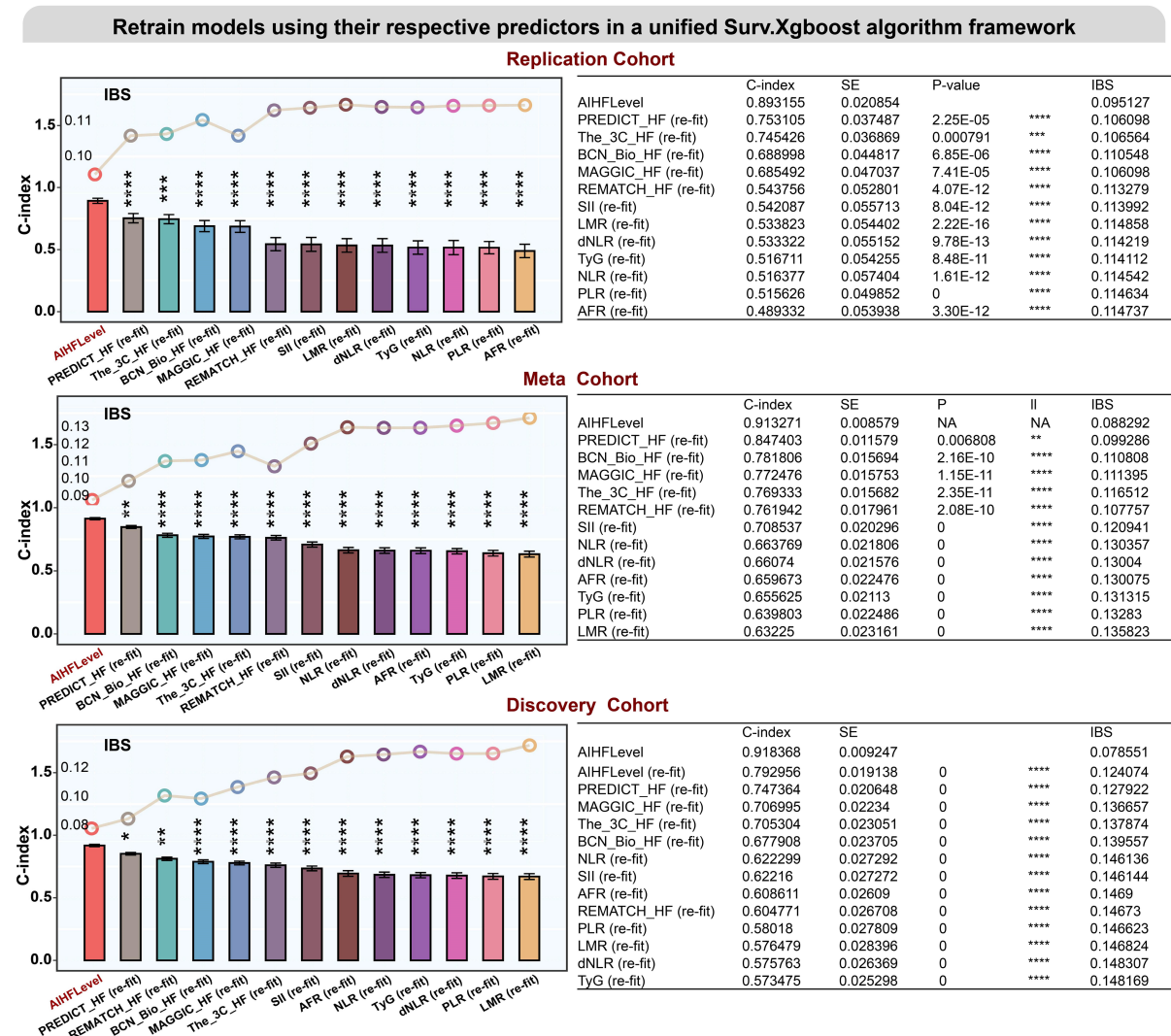
Retrain models using their respective predictors in a unified proportional hazards regression framework



Revised Supplementary Fig. 5

(b) In addition, we employed the Surv.Xgboost algorithm—a key component in AIHFLevel's development—to refit each comparative model using only their respective predictors. This approach established a uniform framework for assessing the inherent strength of each model's selected features in a consistent algorithmic environment. AIHFLevel still maintained superior accuracy across all cohorts (**Revised Supplementary Fig. 6**). The consistent superiority of AIHFLevel across various algorithmic frameworks underscores the inherent robustness and predictive reliability of its selected predictors.

This performance highlights the effectiveness of our AI hybrid modeling framework in identifying a set of robust predictors. The 12-predictor AIHFLevel not only accurately forecasts the prognosis of AHF&RD patients but also does so with a streamlined feature set, greatly enhancing its clinical utility and readiness for broader implementation.



Revised Supplementary Fig. 6

The revised content mentioned above has been incorporated into the 'Robustness and superior performance of AIHFLevel' subsection of the Results section (**revised manuscript: line 189-236, page 8-9**), as well as the Discussion section in the (**revised manuscript: line 467-503, page 17-19**).

We express our heartfelt gratitude for your valuable advice, which we believe will profoundly strengthen the rigor, credibility, and quality of our research. Your insightful suggestions have provided us with a clearer direction and have enriched our study.

Your support and understanding are highly appreciated, and we earnestly hope that our revised manuscript will meet your esteemed approval!

Q4: This site is not accessible <http://www.chictr.org>, so we won't be able to track important aspects of the study, e.g. ethnics

Response:

Thank you once again for bringing this to our attention. We sincerely apologize for any confusion caused by the previous oversight in our manuscript.

In response to your concern, we have taken immediate steps to ensure that all relevant aspects of our study, including ethical considerations, are fully transparent and accessible.

We have updated the manuscript to provide alternative links (<http://www.chictr.org.cn>) and additional information to facilitate the tracking of important study details.

The revised content mentioned above has been incorporated into the 'Data availability' section (**revised manuscript: line 1014-1029, page 36-37**)

The in-house data in this study was accessed from the electronic health records system of the Henan Province Clinical Research Center for Cardiovascular Diseases. This study was conducted to the ethical guideline of the Declaration of Helsinki and approved by the Institutional Ethics Committee of Henan Province Clinical Research Center for Cardiovascular Diseases (No.2021-KY-0720). The study was registered at the Chinese Clinical Trial Registry (<http://www.chictr.org.cn>, No. ChiCTR2200060633). The external data was accessed from the Beth Israel Deaconess Medical Center Resource, which is available at Multi-parameter Intelligent Monitoring III database. We completed the National Institutes of Health Course in the United States, passing the exam of human protection research participants (No.9971167). Moreover, access to the data was granted following the completion of the Collaborative Institutional Training Initiative (CITI) program “Data or Specimens Only Research” (certification number: 41407001). Any additional information required to reanalyze the data reported in this paper is available from the corresponding author contact upon request.

Your diligence in reviewing our work is greatly appreciated, and we hope that these revisions address your concerns satisfactorily.

Q5: Text sizes in some figures are too small (or low resolution) to read, e.g. Figure S1 and Figure S2. So I can't read these results and make judgements.

Response:

Thank you very much for your insightful recommendations.

We have also embellished and restructured the manuscript, figure, and Supplementary figures to improve its readability and logical flow. Detailed information and raw data related to the analysis results have been added to enhance the reproducibility of the research data, and are now included in the **revised Supplementary Tables 1-20**.

Every single figure and Supplementary figure have been polished and restructured for clarity and readability in the **revised Figures 1-7 and revised Supplementary Figures 1-10**.

We sincerely hope that these revisions will meet your approval!

Q6: Legends of most figures only give a title without giving the details. Very difficult to track what the plots are exactly showing

Response:

Thank you for pointing out this issue regarding the clarity of our figure legends.

In response to your feedback, we have revised the legends of all figures in the manuscript to include comprehensive descriptions of what each plot represents, as well as any relevant statistical information or annotations (**Revised Figures 1-7 Legend, Revised manuscript line 1347-1609, pages 45-53**).

We hope that these enhancements will facilitate a better understanding of the visual data presented and contribute to the overall readability of our study.

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

The authors have addressed all the comments in great detail with additional analyses and data. The manuscript is significantly improved.

Reviewer #2 (Remarks to the Author):

I have no further comments

Reviewer #3 (Remarks to the Author):

Q1. I think there may be a misunderstanding. I have no doubts about the importance of the research question of risk prediction for the comorbidity of AHF with RD. However, I would appreciate definitive clarification on the methods used in this study. Given that composite/prognostic prediction methods have been widely used in studies of risk prediction for other disease/comorbidities, whether this study focuses more on applying established methods/frameworks, or if it introduces new methods. Could you clarify on what is novel about the methods proposed in this study if it's latter?

Q2. The word "latent" is misleading/misused in the term "latent survival feature". Obviously, you meant specific clinical variables rather than a latent value from methods like matrix factorization, etc.

Q3. In Figure 3c, how is the AIHFLevel compared with a single machine learning method (e.g. elastic net) from the 12 applied methods on the same set of variables in the external validation? It needs to be clearer whether the composite method plays a major role in improving the performance or simply the addition of clinical features compared with other scores/models.

Point to Point Response

Manuscript Number: NCOMMS-23-44173B

Title: **AI hybrid survival assessment system for advanced heart failure with renal dysfunction**

Reviewer's comment: Black

Authors' responses: Blue

Revised text in the re-submitted files: Red

REVIEWER COMMENTS

Reviewer #3 (Remarks to the Author):

Q1: I think there may be a misunderstanding. I have no doubts about the importance of the research question of risk prediction for the comorbidity of AHF with RD. However, I would appreciate definitive clarification on the methods used in this study. Given that composite/prognostic prediction methods have been widely used in studies of risk prediction for other disease/comorbidities, whether this study focuses more on applying established methods/frameworks, or if it introduces new methods. Could you clarify on what is novel about the methods proposed in this study if it's latter?

Response:

Thanks very much for your continued engagement with our manuscript and for your insightful suggestions that help clarify the focus and contributions of our study. We appreciate the opportunity to further elucidate the methods used in our research and the novel aspects of our approach.

Regarding your query about the methods used in our study, we hope the following supplement provides a clearer explanation:

Our study not only applies existing risk prediction learning algorithms but innovatively developed a new ensemble hybrid model framework (**AI hybrid modeling framework**). This represents an advancement and practical value in the field of risk prediction for AHF with RD comorbidity.

(1) Research Objective and Methodological Innovation Points:

Instead of cherry-picking an learning algorithm, our approach focuses on constructing ensembles to improve survival assessment accuracy, especially for the comorbidity of heart failure with renal dysfunction. We employed 12 different AI learning algorithms and expanded possible model schemes through assembling three distinctive feature engineering approaches (Embedded, Filter, Wrapper), ensuring our model maintains robust generalizability across various clinical scenarios.

Our research draws upon established concepts in AI modeling, specifically the hybrid paradigm that incorporate ensemble-based approaches. This concept has been validated by various researchers who demonstrated varying performances across heart disease risk datasets (*IEEE Trans NanoBioscience*, Dec. 2017;16(8):708–17; *IEEE Access* 2018;6:9256–61; *Improving the Accuracy of Prediction of Heart Disease Risk Based on Ensemble Classification Techniques. Informatics in Medicine Unlocked*, Jan; *IEEE Access* 2019;7:54007–14). Indeed, while AI learning methods have been extensively applied across various medical disciplines, there is a continuous search for optimization and enhancement of these methodologies. Ensemble learning, in particular, is a robust approach that significantly augments the performance of machine learning tasks by leveraging the collective strengths of multiple models (*IEEE Transactions on Neural Networks and Learning Systems*, Dec. 2017; 28(12):2985–97). Moreover, there is growing advocacy for the use of hybrid learning models—particularly those that integrate ensemble-based methods, combining two or more models to enrich the existing learning frameworks. Mienye and colleagues have demonstrated that ensemble-based learning can outperform individual models in predicting heart failure events, substantiating the effectiveness of this approach (*An Improved Ensemble Learning Approach for the Prediction of Heart Disease Risk. Informatics in Medicine Unlocked*). Ensemble modeling enhances the prediction process by integrating a diverse set of independent models, which helps in reducing generalization errors and bolstering the reliability of predictive outcomes (*AMIA Annu Symp Proc*, PMID: 30815108).

This foundational understanding informs the design of our novel AI hybrid modeling framework, which we employ to optimize the predictive accuracy and applicability of our models in clinical settings.

(2) Uniqueness of the Technical Framework:

Compared to traditional single models or simple combinations of a few models, our AI hybrid modeling framework integrates multiple independent models for predictions, significantly reducing generalization errors. This framework includes the following features:

(a) Integration of Learning Algorithms: Integration of 12 classic, proven, or superior learning algorithms, generating 15 schemes: L1.regularization (lambda.min), L1.regularization (lambda.1se), L2.regularization (lambda.min), L2.regularization (lambda.1se), ENet.regularization (lambda.min), ENet.regularization (lambda.1se), Surv.RFsrc, Surv.SVM, Surv.rpart, Surv.Coxboost, Surv.glmboost, Surv.Xgboost, Surv.gbm, Surv.Fully-parametric,

Surv.aorsf. This diversity allows for robust generalizability and optimized prediction accuracy across different clinical scenarios.

(b) Feature Engineering Approaches: Ensemble methods utilizing Embedded, Filter, and Wrapper approaches: regularization, sequential forward floating selection (SFFS), and Correlation-Adjusted Survival Score (CARS) to explore a broad spectrum of modeling schemes, thereby enhancing the predictive capability of our framework.

(c) Rigorous Assessment Strategy: A systematic and unbiased assessment strategy including 10 repeated 10-fold cross-validation, Monte-Carlo cross-validation (MCCV), and bootstrap analysis. This strategy enhances the model's generalizability and reliability in diverse clinical settings and populations.

(3) : Robust AIHFLevel generated from this framework

The survival assessment system AIHFLevel, produced through this framework, shows superior and stable performance with extrapolative capabilities. This confirms previous researchers' views that a stacked ensemble can provide an easily extensible and adaptable framework for integrating multiple component systems, maximizing the efficacy and advantages of different learners.

In summary, our framework not only applies but significantly builds upon and extends existing methodologies in predictive modeling, presenting a novel and effective approach to addressing the complexities of heart failure with renal dysfunction. This methodological innovation is central to our contribution to the field, enhancing the predictive accuracy and clinical utility of risk assessments in this challenging area of healthcare.

The revised content mentioned above has been incorporated into the 'Discussion' section (revised manuscript: line 450-477, page 17-18).

Indeed, our study not only applied existing risk prediction learning algorithms but innovatively developed a novel ensemble hybrid model framework, potentially marking a significant advancement in the field of risk prediction for AHF with RD comorbidity. We draw upon established concepts in AI modeling, notably hybrid paradigms that incorporate ensemble-based approaches.^{52, 53} Herein, instead of choosing the best individual learning algorithm, our methodology focuses on constructing ensemble modeling schemes to improve survival assessment accuracy, especially for the comorbidity of AHF with RD. We employed 12 different AI learning algorithms and broadened potential model schemes by integrating three distinctive feature engineering approaches: Embedded, Filter, and Wrapper. This integration reduces generalization errors and ensures robust generalizability across various clinical scenarios. The novel of our framework methodology included: (1) Integration of multiple learning algorithms: We incorporate a diverse array of advanced algorithms, each contributing uniquely to the ensemble. This integration enhances robustness and reduces the likelihood of model overfitting. Included are various forms of regularization and sophisticated machine learning techniques such as Support Vector Machines, Random Forests, and Gradient Boosting Machines. (2) Advanced feature engineering techniques: Our model employs a combinational family of regularization, sequential forward floating selection (SFFS), and Correlation-Adjusted Survival Score (CARS). This allowed for a comprehensive exploration of potential predictive variables and enhances the model's ability to handle high-dimensional data effectively. (3) Ensemble-based modeling: Leveraging the strengths of ensemble learning, our framework boosts predictive accuracy by strategically combining models to mitigate individual weaknesses and capitalize on their collective strengths, thus ensuring more reliable and consistent predictions. (4) Comprehensive assessment strategy: A systematic and unbiased assessment strategy included 10 repeated 10-fold cross-validation, Monte-Carlo cross-validation, and bootstrap analysis, which enhanced the model's generalizability and reliability in diverse clinical settings and populations.

Once again, we appreciate your insights, which have greatly assisted in refining our research and methodology. We look forward to your further suggestions and approval.

Q2: The word “latent” is misleading/misused in the term “latent survival feature”. Obviously, you meant specific clinical variables rather than a latent value from methods like matrix factorization, etc.

Response:

Thank you for your astute observation regarding the use of the term “latent” in describing the survival features identified in our study. Upon reflection and consideration of your feedback, we agree that the term may inadvertently suggest a methodological context that does not align with how these features were derived.

To clarify, the 46 variables initially identified through Kaplan-Meier and Cox proportional hazards analysis were preliminary and exploratory, serving as potential predictors for further analysis. Thus, the term “candidate survival features” is indeed more appropriate, as it better represents their role in the developmental phase of our predictive modeling process.

Therefore, we revised our manuscript to replace “latent” with “**candidate**” for the initial set of variables and ensure that the distinction between these candidate features and the significant features used in the final model is clear and precise.

We appreciate your insightful feedback, which has undoubtedly helped enhance the accuracy and clarity of our manuscript.

Q3: In Figure 3c, how is the AIHFLevel compared with a single machine learning method (e.g. elastic net) from the 12 applied methods on the same set of variables in the external validation? It needs to be clearer whether the composite method plays a major role in improving the performance or simply the addition of clinical features compared with other scores/models.

Response:

When comparing our AIHFLevel with other models, we expanded our analysis to include 12 published risk indicators and models, such as the systemic immune-inflammation index (SII), neutrophil-lymphocyte ratio (NLR), neutrophils to leukocyte-neutrophil count (dNLR), lymphocyte-monocyte ratio (LMR), platelet-lymphocyte ratio (PLR), albumin-to-fibrinogen ratio (AFR), triglyceride-glucose (TyG) index, MAGGIC-HF, PREDICT-HF, BCN Bio-HF, REMATCH-HF, and the 3C-HF score.

Considering your question about how the modeling method of AIHFLevel compares to those of these 12 models when using the same variables, we encountered the following issue: these well-known models typically generate a prognostic score by multiplying several variables by corresponding coefficients, or by integrating variables into a more complex formula. This means that there is no specific modeling method per se. Upon further review of the literature on these models, we found that most employed the Cox proportional hazards model combined with regularization techniques during their development. This allowed us to add a comparative model by refitting the variables used in AIHFLevel using the Cox proportional hazards method to create a prognostic model for comparison with AIHFLevel. This refitted model was reflected in **Figure 3c**, We have added this detail in the revised manuscript. (**revised manuscript: line 212-217, page 9**).

Therefore, **in the first comparison**, AIHFLevel is contrasted against the 12 models (each using their predictive variables and original formulas or coefficients) and a Cox model (a prognostic model refitted using the Cox regression method with AIHFLevel variables).

In the second comparison, presented in **Supplementary Fig. 5**, we engaged in a refitting process where AIHFLevel and the other 12 models were retrained using only their predictive variables based on the proportional hazards approach, discarding their original coefficients or

formulas. This methodological improvement was aimed at facilitating a fair, standardized comparison, thus validating the robustness of AIHFLevel in a more stringent evaluative setting.

In the third comparison, shown in **Supplementary Fig. 6**, we also utilized the Surv.Xgboost algorithm—which is the primary modeling method of AIHFLevel—to refit each comparative model using only their respective predictive variables. This approach established a uniform framework for assessing the inherent strength of each model's selected features in a consistent algorithmic environment.

As a result, in all three comparisons, across all cohorts, AIHFLevel consistently maintained higher accuracy. Based on the above analyses, we found that AIHFLevel consistently outperforms across various algorithmic frameworks, highlighting the inherent robustness and predictive reliability of its selected predictors.

Nevertheless, as you mentioned, since we do not have access to the specific operations undertaken by the developers of each model during their development, such as what feature engineering was applied or the specific parameters of the risk ratio regression models used, AIHFLevel cannot truly compare methodologies with each model. **We believe this represents a shortcoming of this study, and we added this limitation to the manuscript. (revised manuscript: line 533-544, page 20).**

However, as described above, we opted for a secondary approach—refitting a prognostic model using the Cox proportional hazards method with AIHFLevel variables for comparison, which unsurprisingly demonstrated that, even using the same set of variables, the modeling method of AIHFLevel still prevails.

For your second question, the results of the three comparisons may serve to illustrate:

(1) When using the same set of predictive variables, the modeling method of AIHFLevel prevails;

(2) When employing the same algorithmic modeling framework, the predictive variables of AIHFLevel prevail.

The consistently superior performance of AIHFLevel further indicates its intrinsic robustness and reliability in predictions. These findings affirm the efficacy of our AI hybrid modeling framework in identifying a robust set of predictors. Our 12-predictor AIHFLevel not only accurately predicts the prognosis for patients with AHF and RD but also achieves this with a streamlined set of features, greatly enhancing its clinical usability and readiness for broader application.

We appreciate the time and effort you have invested in helping us improve our work. We hope that our work meets your expectations and contribute positively to the field.

REVIEWERS' COMMENTS

Reviewer #3 (Remarks to the Author):

I don't have any further comments. The authors have addressed my questions.